



GE Medical Systems

gehealthcare.com

Technical Publication

**Direction 2301052-100
Revision 01**

**GE Medical Systems
UPS Installation: HSA & LS1.X-4.X**

The information contained in this manual applies to the following CT Power Distribution Units (PDU):

- CT Global PDU
- Compact PDU
- H2 Compact PDU
- NGPDU-2 (2326492-2 only)

As used on the following CT Systems:

- HSA-RP
- HS CT/I
- LightSpeed 1.x
- LightSpeed 2.x
- LightSpeed 3.X
- LightSpeed 4.X



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- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	09/05/01	FPR # 1006001 Initial Document Release
1	12/20/04	PQR 13015157

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Chapter 1

Common Instructions and Requirements

Section 1.0

Scope

This installation specification is a control document for the installation of an Uninterruptible Power System (UPS) unit to be used by General Electric Medical Systems (GEMS) in the Computerized Tomography diagnostic imaging system. This document supports CT Systems that use one of the following Power Distribution Units (PDU):

- 2113764 Global PDU
- 2133533 CT Compact PDU
- 2269902 CT (H2) Compact PDU
- 2326492-2 NGPDU (Note: Only the “-2” model is supported)

It also includes the connection instructions for a Control Panel and a Disconnect Panel to be installed with the UPS unit.

Section 2.0

Overview

This document was designed to allow installation of a 12kVA UPS into CT/RP 2.X, CT/i, LightSpeed QXI, LightSpeed Plus and LightSpeed Ultra systems using the Global, Compact and Compact (H2) PDUs. The UPS primarily backs up the system computer functions. It is intended to bridge short power outages and provide time for crossover from normal mains power to emergency power. It is sized to provide 10 minutes backup with a 12kVA load with fully charged batteries. At the normal 6 kVA load with fully charged batteries the time will of course exceed 10 minutes. It is recommended that if a power outage, with no mains or emergency power, exceeds 5 minutes, the patient should be removed from the table and a shutdown of the system should be initiated.

The instruction presented in this document includes two different UPS model: POWERWARE and LIEBERT UPS.

A Control Panel is also part of the UPS unit. This panel primarily provides interlocking that allows safe and orderly system startup and shutdown.

Finally, a Disconnect Panel is installed with the UPS unit to provide electrical disconnect of the UPS from the CT System. It also provides a contact closure that shuts the UPS down when system power is turned off by way of any of the emergency off switches.

Section 3.0

Safety Requirements - General Warning Considerations

The only user operations permitted with the UPS are:

- Connection of the UPS to the CT PDU.
- Starting up and shutting down of the UPS.
- Operating the user interface in the UPS.
- Connecting data interface cables.
- Monitoring the UPS.

These operations must be performed according to the instruction in this document and the User's Guide for the UPS system been installed.

Note: It is strongly recommended to have the User's Guide for the particular UPS been used. This installation document covers the POWERWARE 9150 and the LIEBERT UPS only.



DANGER DANGER THE UPS CONTAINS LETHAL VOLTAGES. ALL REPAIRS AND SERVICES SHOULD BE PERFORMED BY AUTHORIZED SERVICE PERSONNEL ONLY.

THERE ARE NO USER SERVICEABLE PARTS INSIDE THE UPS.

Section 4.0

Preliminary Instructions

4.1 Electrician

An electrician will be required to mount the UPS Control Panel next to the A1 Disconnect Panel. A 3/4" knockout is provided on the right side of the UPS Control Panel for wiring to the A1 Disconnect Panel. A 1/2" conduit¹ will also be required between the A1 Disconnect and the computer floor or to the general area of the UPS, for the control cable going to the UPS Disconnect Panel, with a box at the end.

4.2 Uncrating

Locate the uncrating instructions for the UPS found on or in the UPS packing. Uncrate the UPS per these instructions. Visually inspect the unit for shipping damage and battery seating. The POWERWARE 9150 model requires assembly of the subsystem. Accomplish this per drawing 110719345 in the POWERWARE 9150 (Physical manual supplied with the unit).

¹.Conduit is a Metal pipe used for panel to panel wiring

4.3 Contents of UPS Interface Kit for GPDU (2143160)

QTY	NAME	DESCRIPTION																																	
1	2139965	UNINTERRUPTIBLE POWER SUPPLY																																	
1	2143161	GPDU UPS DISCONNECT PANEL																																	
1	2143207	GPDU UPS CONTROL PANEL																																	
1	2147944	HARDWARE KIT FOR GPDU UPS:																																	
		<table> <tr> <th>Qty</th><th>Name</th><th>Description</th></tr> <tr> <td>1</td><td>2139051</td><td>UPS INPUT POWER CABLE-PDU TO UPS</td></tr> <tr> <td>1</td><td>2139056</td><td>UPS OUTPUT PWR CBLE CONTACT TO PDU</td></tr> <tr> <td>1</td><td>2139059</td><td>CONTACTOR ASM TO UPS INTERLOCK PANEL</td></tr> <tr> <td>1</td><td>2142365</td><td>UPS DISCONNECT BOX MOUNTING BRKT</td></tr> <tr> <td>1</td><td>2142363</td><td>LEIBERT UPS/REAR OUTPUT COVER PLATE</td></tr> <tr> <td>1</td><td>2142579</td><td>RHAP UPS LEAD DATA CHART</td></tr> <tr> <td>1</td><td>2139057</td><td>EXTENSION CBL-CONTACTOR ASM TO UPS</td></tr> <tr> <td>1</td><td>2301052-100</td><td>UPS INSTALLATION: HSA & LS1.X-4.X</td></tr> <tr> <td>1</td><td>2142725</td><td>GLOBAL PDU/ADAPTER PLATE</td></tr> <tr> <td>1</td><td>2139060</td><td>CONTACTOR ASM TO UPS REPO</td></tr> </table>	Qty	Name	Description	1	2139051	UPS INPUT POWER CABLE-PDU TO UPS	1	2139056	UPS OUTPUT PWR CBLE CONTACT TO PDU	1	2139059	CONTACTOR ASM TO UPS INTERLOCK PANEL	1	2142365	UPS DISCONNECT BOX MOUNTING BRKT	1	2142363	LEIBERT UPS/REAR OUTPUT COVER PLATE	1	2142579	RHAP UPS LEAD DATA CHART	1	2139057	EXTENSION CBL-CONTACTOR ASM TO UPS	1	2301052-100	UPS INSTALLATION: HSA & LS1.X-4.X	1	2142725	GLOBAL PDU/ADAPTER PLATE	1	2139060	CONTACTOR ASM TO UPS REPO
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1	2139060	CONTACTOR ASM TO UPS REPO																																	

Table 1-1 UPS Interface Kit for GPCU (2143160)

4.4 Electrical Requirements - Primary Input Power

Input Voltage Range: 120/208 VAC 120° phase displacement.

Frequency: 50 / 60 Hz (47 to 53 or 57 to 63) Hz.

Chapter 2

Powerware 9150 UPS Installation

Perform the following installations and connections at the UPS and UPS Disconnect Panel.

Section 1.0

Powerware 9150 UPS Input Connections

Refer to Sheet 1 of 2142192IDW, on [page 82](#).

1.1 Cover Removal

- 1.) Remove the front and left side covers from the POWERWARE 9150 UPS as follows:
 - a.) The front panel is held in place by spring clips. To remove the front panel, pull front panel forward using the “Hand hold” provided on the lower right side of the front panel.
 - b.) Two screws (top and bottom) should be removed from the lower left corner of the assembly to release the Left Panel.
 - c.) The left side panel is removed by pulling the front edge of the left panel outward and then by sliding the panel towards the back of the UPS.
- 2.) Verify that all of the battery breakers on front of UPS are in the “Off” position.
- 3.) Confirm by measurement that all terminals on terminal strip are at ground potential.

REPLACE THE REAR COVER AS FOLLOWS:

- 1.) Remove and discard rear cover panel.
- 2.) Retain six (6) mounting screws.
- 3.) Install replacement rear panel from kit using six (6) retained mounting screws.

1.2 UPS / Disconnect Panel

Install UPS / Disconnect Panel as follows:

- 1.) Assemble a 3/4” conduit fitting in bottom of disconnect box, using mounting hardware supplied in the kit.
- 2.) Assemble a 1/2” conduit fitting in bottom of disconnect box, using mounting hardware supplied in the kit.
- 3.) Mount UPS disconnect box on UPS rear cover panel, using four (4) mounting studs and hardware supplied.

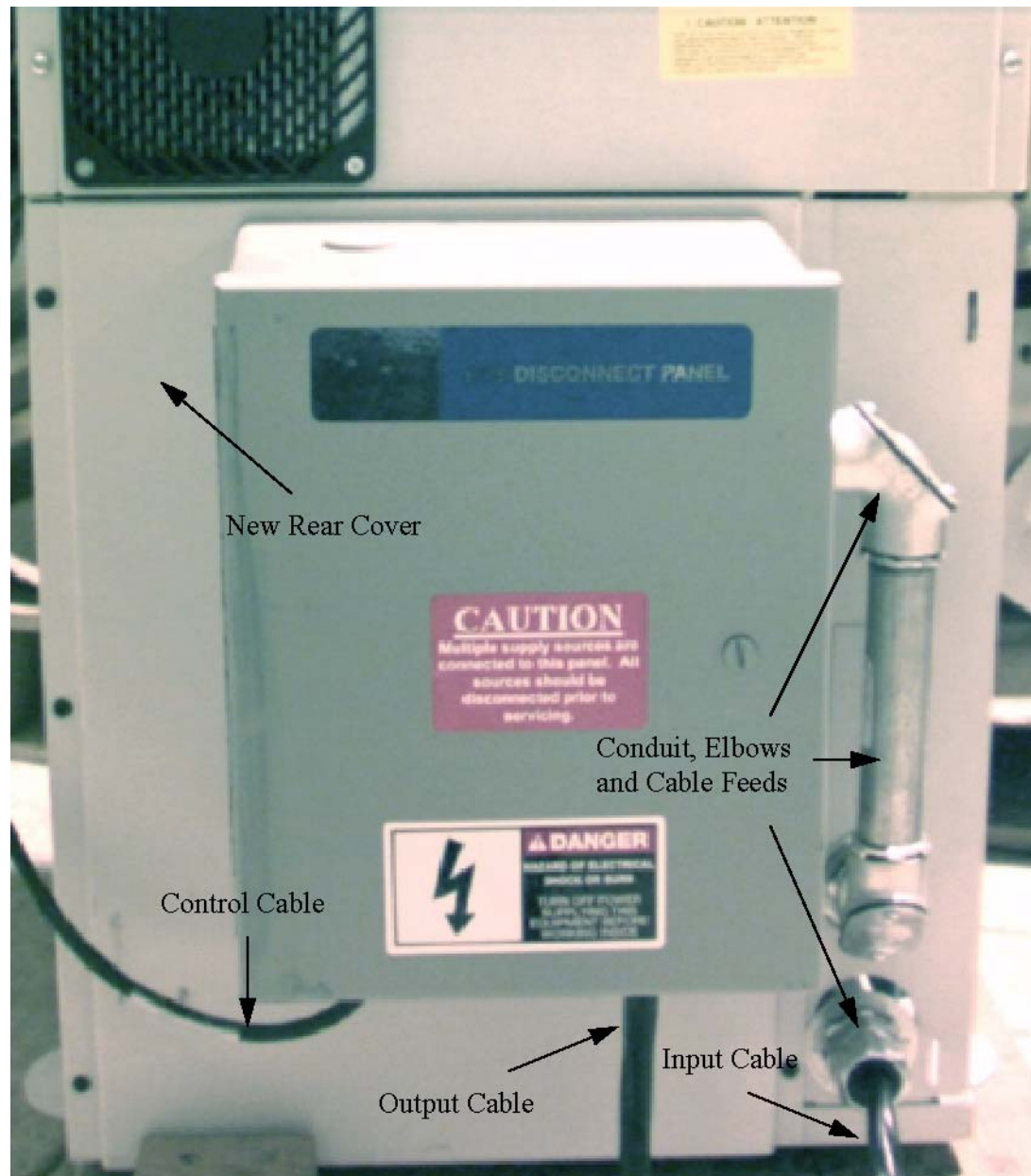


Figure 2-1 UPS Disconnect Panel

1.3 UPS / Disconnect Panel wiring conduit.

Install UPS / Disconnect Panel wiring conduit as follows:

- 1.) Assemble 3/4" conduit elbow to one end of 6" conduit using lock ring supplied. Assemble second 3/4" conduit elbow to other end of the 6" conduit using lock ring.
- 2.) Assemble conduit sub-assembly to the right side of the disconnect box using hardware supplied in kit. Assemble sub-assembly to the UPS using adapters and hardware supplied in kit.
- 3.) Remove both conduit elbow cover plates and retain for re-assembly after wiring is in place.
- 4.) Install 3/4" conduit in remaining large hole below the conduit assembly using a short nipple, adapter washers and mounting hardware supplied in the kit.
- 5.) Tighten all mounting and lock rings.

1.4 UPS Disconnect Panel Input Cable from PDU

Connect Input cable as follows, Reference [Figure 2-1 on page 24](#) and Reference [Figure 2-2 on page 25](#).

- 1.) Pull input cable (P/N 2139051) through the 3/4" conduit in the back of the UPS, to TB1.
- 2.) Connect the GREEN/YELLOW wire to TB1-1 (GND).
- 3.) Connect the RED wire to TB1-2 (L1).
- 4.) Connect the BLACK wire to TB1-3 (L2).
- 5.) Connect the BLUE wire to TB1-4 (NEUT).

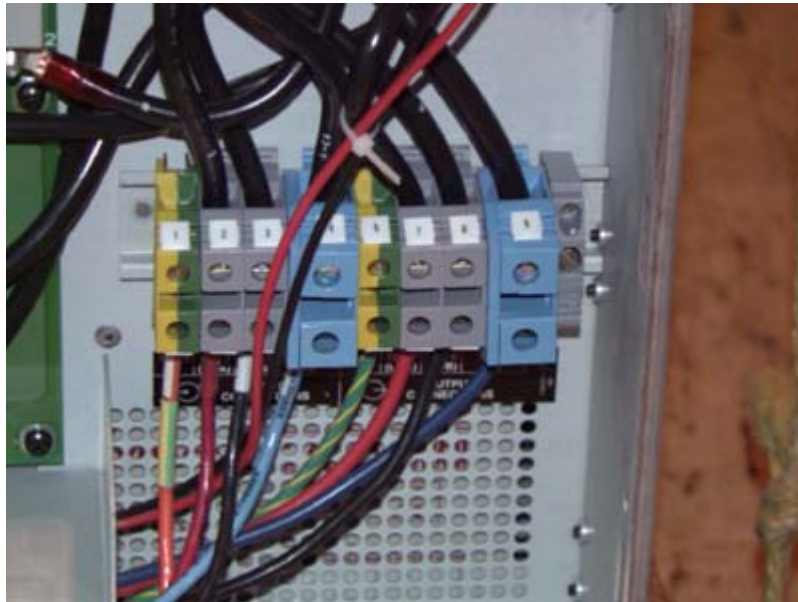


Figure 2-2 UPS 9150 TB1 Terminal Block

Section 2.0 Powerware 9150 UPS Output Connections

Refer to Sheet 1 of 2142192IDW, on [page 82](#).

2.1 Output of the UPS to UPS Disconnect Panel

On the left side of the UPS, locate the UPS output Data Lead (P/N 2139057) in the kit and install as follows: Reference [Figure 2-2 on page 25](#).

- 1.) Connect the GREEN/YELLOW (#10) wire to TB1-6 (GND).
- 2.) Connect the RED (#10) wire to TB1-7 (L1).
- 3.) Connect the BLACK (#10) wire to TB1-8 (L2).
- 4.) Connect the BLUE (#10) wire to TB1-9 (Neutral).

Note: TB1-4 and TB1-5 are tied together internally (Input Neutral) and TB1-9 and TB1-10 are tied together internally (Output Neutral).

2.2 UPS Internal Interlock (REPO)

Connect the interlock as follow: Reference [Figure 2-3 on page 26](#).

- 1.) Locate the 2 pin male connector (REPO) plugged into the bottom side of the upper front panel. Un-plug the connector and remove the jumper, which connects the two pins.
- 2.) Connect the RED (#16) and BLACK (#16) wires from the Lead Data kit to the two terminals of the connector.
- 3.) Replace the connector in its socket.



Figure 2-3 UPS 9150 Internal Interlock (REPO)

2.3 REPO Cable Routing

- 1.) Dress black and red leads down to the TB1 area so that covers will not pinch them.
- 2.) Dress all six (6) wires from section 2.1 and section 2.2, across bottom of the UPS, through the conduit assembly and into the UPS disconnect box.
- 3.) Replace both conduit elbow access covers.
- 4.) Replace UPS left side cover and secure with single screw.

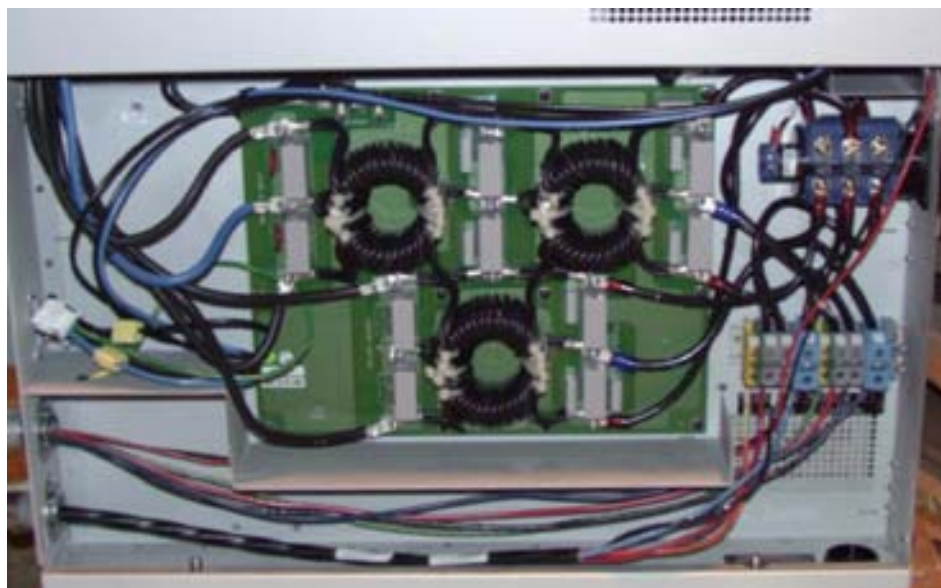


Figure 2-4 UPS 9150 Left Side Interior

Section 3.0

Powerware UPS Disconnect Panel Connections

Make the following connections in the UPS Disconnect Panel as described below.
Refer to Sheet 1 of 2142192IDW, on [page 82](#).

Note: These instructions are applicable for both model of POWERWARE UPS (9150 and PLUS) described in this document.

3.1 UPS Disconnect Panel Wiring

Note: Steps 1 through 4 cable are from UPS TB1-6,7,8,9. Steps 5 and 6 are REPO wires.

- 1.) Connect the GREEN/YELLOW (#10) wire to terminals GND.
- 2.) Connect the BLUE (#10) wire to terminal NEUT.
- 3.) Connect the BLACK (#10) wire to lower connection in CB D30.
- 4.) Connect the RED (#10) wire to upper connection in CD D30.
- 5.) Connect the BLACK (#16) wire from the REPO to 13NO on the top of M2 contactor (44 NO).
Reference [Figure 2-5 on page 28](#) or [Figure 2-6 on page 29](#).
- 6.) Connect the RED (#16) wire from the REPO to 14NO on the top of M2 contactor (43 NO).
Reference [Figure 2-5 on page 28](#) or [Figure 2-6 on page 29](#).

Note: If the REPO circuit is wired to the wrong contacts in the Disconnect Panel, the power at the Main A1 panel won't latch.

3.2 UPS Disconnect Panel Control Cable to UPS Start/Stop Control Panel

Pull control cable (P/N 2139059 coming from control panel) through 1/2" conduit at bottom of UPS disconnect panel and connect to UPS Start/Stop panel as follows:

- 1.) Connect the GREEN/YELLOW (#16) wire to terminal GND.
- 2.) Connect the BLACK (#16) wire to terminal F1.
- 3.) Connect the RED (#16) wire to the M2 Coil terminal.
- 4.) Connect the BLUE (#16) wire to the Neutral terminal.
- 5.) Tighten strain relief onto cable.

3.3 UPS Disconnect Output Cable to PDU

Pull UPS Output cable (P/N 2139056) going to the PDU through the 3/4" conduit in bottom of UPS disconnect panel and connect as follows:

- 1.) Connect the GREEN/YELLOW (#10) wire to GND.
- 2.) Connect the BLUE (#10) wire to neutral (NEUT).
- 3.) Connect the RED (#10) wire to terminal 4T2 on the M2 contractor.
- 4.) Connect the BLACK (#10) wire to terminal 2T1 on the M2 contractor.
- 5.) Tighten strain relief onto cable.

Proceed to the appropriate PDU Chapter for the next steps of installation.

The A1 to UPS assembly, Interlock Cable (P/N 2142579) and Control Cable (P/N 2139059) will be installed in [Chapter 8 - A1 Disconnect Panels](#).

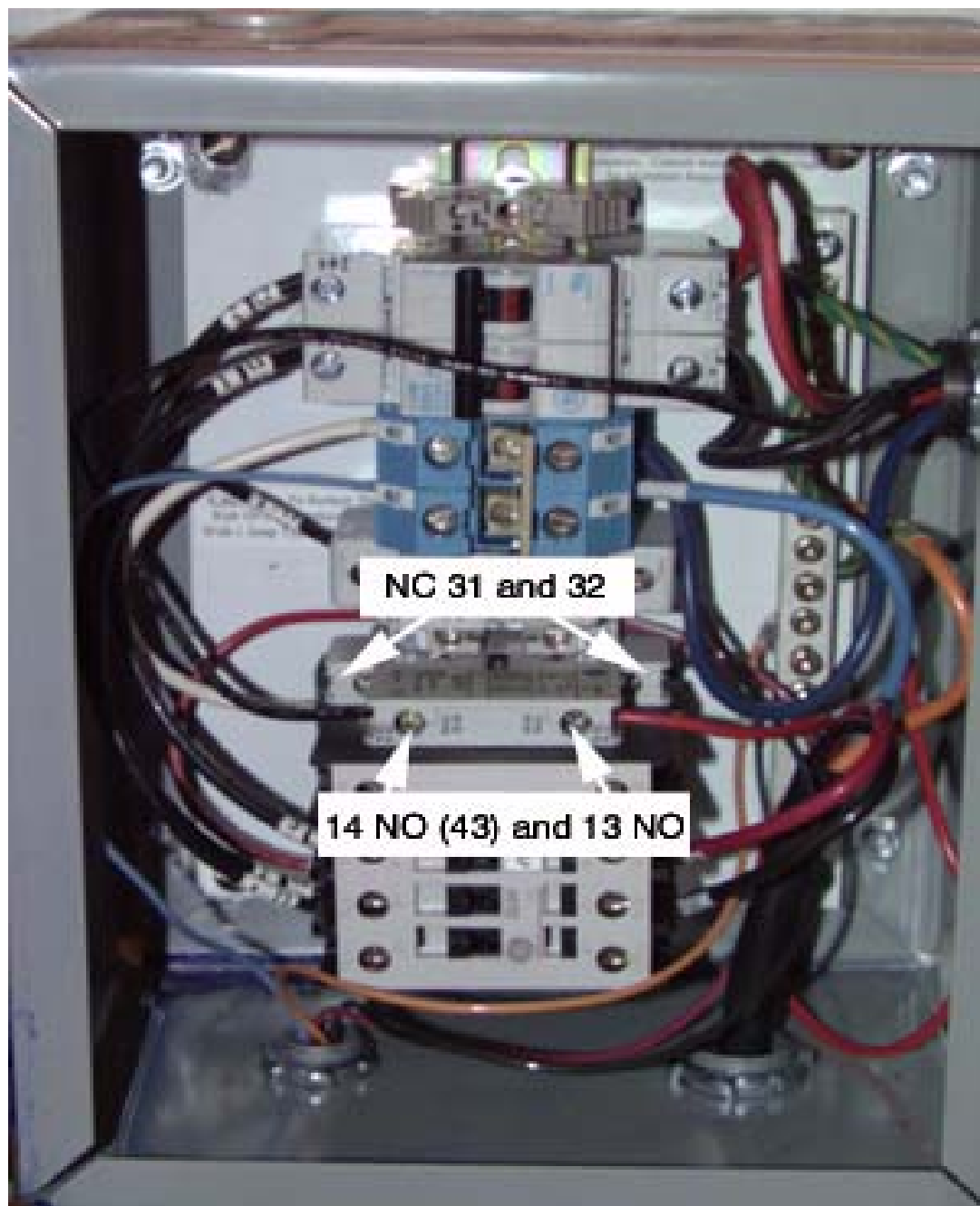


Figure 2-5 UPS Disconnect Panel Interior

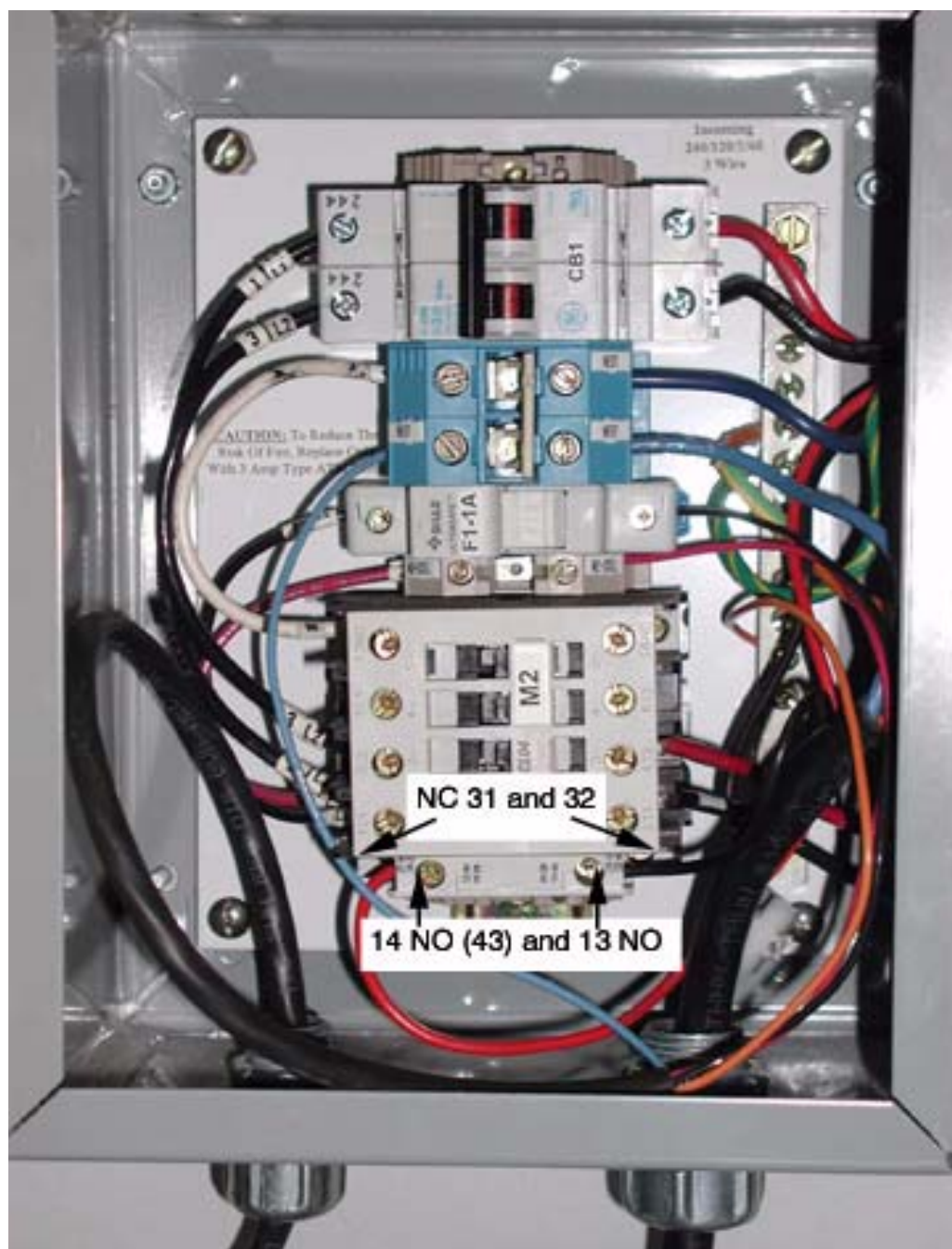


Figure 2-6 Earlier model UPS Disconnect Panel

Chapter 3

Installation for Powerware 12/12 UPS

Section 1.0

Powerware Plus Model 12/12 UPS Input Connections

Refer to Sheet 1 of 2142192IDW, on [page 82](#), and follow these instructions when applicable. Notice from the POWERWARE PLUS User's Guide that this UPS unit physically differs from the 9150 unit.

Note: The Powerware Plus model 12/12 is no longer available. The information in this section is provided for historical purposes and as an aid in re-installing an existing UPS. Depending upon the age of the UPS, some of the terminal markings may differ from those specified here. Where discrepancies exist, make corrections to the labeling as required to implement the functionality shown in schematic 2142192SCH.

1.1 UPS / Disconnect Panel

Install UPS / Disconnect Panel as follows:

- 1.) Assemble a ¾" conduit in bottom of disconnect box, using mounting hardware supplied in the kit.
- 2.) Assemble a ½" conduit in bottom of disconnect box, using mounting hardware supplied in the kit.
- 3.) Mount UPS disconnect box on UPS rear cover panel, using four (4) mounting studs and hardware supplied.

1.2 UPS / Disconnect Panel Wiring Conduit

Install UPS / Disconnect Panel wiring conduit as follows:

- 1.) Assemble ¾" conduit elbow to one end of 6" conduit using lock ring supplied. Assemble second ¾" conduit elbow to other end of the 6" conduit using lock ring.
- 2.) Assemble conduit sub-assembly to the right side of the disconnect box, using hardware supplied in kit. Assemble sub-assembly to the UPS, using adapters and hardware supplied in kit.
- 3.) Remove both conduit elbow cover plates and retain for re-assembly after wiring is in place.
- 4.) Install ¾" conduit in remaining large hole below the conduit assembly using a short nipple, adapter washers and mounting hardware supplied in the kit.
- 5.) Tighten all mounting and lock rings.

1.3 UPS Disconnect Panel Input Cable from PDU

Connect Input cable as follows:

- 1.) Route the input cable (P/N 2139051) through the input connector in the back of the UPS Power Cable Termination Block.
- 2.) Connect the GREEN/YELLOW wire to position 1 (GND).
- 3.) Connect the BLUE wire to position 2 (Neutral).
- 4.) Connect the RED wire to position 4 (L1).
- 5.) Connect the BLACK wire to position 5 (L2).

Section 2.0

Powerware Plus Model 12/12 UPS Output Connections

Refer to Sheet 1 of 2142192IDW, on [page 82](#).

2.1 Output of the UPS

Connect the Output of the UPS (backside). Locate UPS output Lead Data (P/N 2139057) in the kit and install as follows:

- 1.) Remove the bonding jumper between terminals 7 (GND) and 8.
- 2.) Connect the GREEN/YELLOW (#10) wire to position 7(GND).
- 3.) Connect the BLUE (#10) wire to position 11 (Neutral).
- 4.) Connect the RED (#10) wire to position 12 (L1).
- 5.) Connect the BLACK (#10) wire to position 13 (L2).

2.2 UPS Internal Interlock (REPO)

Connect the interlock as follows:

- 1.) Locate the 2 pin male connector (REPO) plugged into the backside of the unit at the Power Cable Termination Block. Unplug the connector and remove the jumper, which connects the two pins.
- 2.) Connect the RED (#16) wire to position 14 and BLACK (#16) wire to position 15, from the Lead Data kit to the two terminals of the connector.
- 3.) Replace the connector in its socket.

Section 3.0

Powerware UPS Disconnect Panel Connections

Make the following connections in the UPS Disconnect Panel as described below. [Refer to Sheet 21](#).

Note: These instructions are applicable for both model of POWERWARE UPS (9150 and PLUS) described in this document.

3.1 UPS Disconnect Panel Wiring

Note: Steps 1 through 4 cable are from UPS TB1-6,7,8,9. Steps 5 and 6 are REPO wires.

- 1.) Connect the GREEN/YELLOW (#10) wire to terminals GND.
- 2.) Connect the BLUE (#10) wire to terminal NEUT.
- 3.) Connect the BLACK (#10) wire to lower connection in CB D30.
- 4.) Connect the RED (#10) wire to upper connection in CD D30.
- 5.) Connect the BLACK (#16) wire from the REPO to 21NC on the top of M2 contactor (32 NO). Reference [Figure 2-5 on page 28](#).
- 6.) Connect the RED (#16) wire from the REPO to 22NC on the top of M2 contactor (31 NO). Reference [Figure 2-5 on page 28](#).

Note: If the REPO circuit is wired to the wrong contacts in the Disconnect Panel, the power at the Main A1 panel won't latch.

3.2 UPS Disconnect Panel Control Cable to UPS Start/Stop Control Panel

Pull control cable (P/N 2139059 coming from control panel) through ½" conduit at bottom of UPS disconnect panel and connect to UPS Start/Stop panel as follows:

- 1.) Connect the GREEN/YELLOW (#16) wire to terminal GND.
- 2.) Connect the BLACK (#16) wire to terminal F1.
- 3.) Connect the RED (#16) wire to the M2 Coil terminal.
- 4.) Connect the BLUE (#16) wire to the Neutral terminal.
- 5.) Tighten strain relief onto cable.

3.3 UPS Disconnect Output Cable to PDU

Pull UPS Output cable (P/N 2139056) going to the PDU through the 3/4" conduit in bottom of UPS disconnect panel and connect as follows:

- 1.) Connect the GREEN/YELLOW (#10) wire to GND.
- 2.) Connect the BLUE (#10) wire to neutral (NEUT).
- 3.) Connect the RED (#10) wire to terminal 4T2 on the M2 contractor.
- 4.) Connect the BLACK (#10) wire to terminal 2T1 on the M2 contractor.
- 5.) Tighten strain relief onto cable.

Proceed to the appropriate PDU Chapter for the next steps of installation.

The A1 to UPS assembly, Interlock Cable (P/N 2142579) and Control Cable (P/N 2139059) will be installed in [Chapter 8 - A1 Disconnect Panels](#).

Chapter 4

Liebert UPS Installation

Do the following installations and connections at the UPS and UPS Disconnect Panel.

Section 1.0

Liebert UPS Input Connections

Make the following connections at the Power Input Terminal Block at the lower center rear of the LIEBERT UPS. It is important to make the LIEBERT connections in the order described here. The terminal block area will be covered by the UPS Disconnect Panel later. Refer to Sheet 2 of 2142192IDW, on [page 83](#).

- 1.) Remove and save the 6 screws in each of the 2 access panels at the lower rear of the UPS.
 - a.) Locate the LIEBERT I/O Panel (P/N 2142363) in the kit for the lower access area and conduit fitting. It has 2 holes punched in it.
 - b.) Install nipples, nut and pull elbows as shown in sheet 22.
 - c.) Mount this panel across the lower access area with the screws saved above.
- 2.) Locate the PDU to UPS Input Cable (P/N 2139051) in the kit and feed it through the left conduit. This is the input connection area. Feed it until the cable outer sheath is just protruding beyond the inside of the conduit and tighten.
 - a.) Connect the cable GREEN/YELLOW wire to the input GND terminal.
 - b.) Connect the BLUE wire to the N terminal.
 - c.) Connect the RED wire to the L1 terminal.
 - d.) Connect the BLACK wire to the L2 terminal.

Section 2.0

Liebert UPS Output Connections

Make the following connections at the Power Output Terminal Block at the lower right rear of the LIEBERT UPS. Refer to Sheet 2 of 2142192IDW, on [page 83](#).



DANGER

THE SPECIAL MODEL LIEBERT UPS SUPPLIED FOR THIS APPLICATION HAS THE NEUTRAL/GROUND BOND ALREADY REMOVED AND DOES NOT HAVE TO BE ACCOMPLISHED IN THE FIELD. VERIFY THAT THE PROPER LIEBERT MODEL UPS (12KVA) HAS BEEN SUPPLIED AS CALLED FOR IN THIS KIT CONTENTS LIST SUPPLIED WITH THIS KIT. USE OF ANY OTHER MODEL COULD RESULT IN A SAFETY HAZARD.

- 1.) Locate UPS Output Lead Data (P/N 2139057) in the kit. Thread it through the 90° elbow in the right opening in the I/O panel installed above. In making the connections, note that the output block is arranged in a quad with the N and G terminals at the top and the L1 & L2 terminals at the bottom.
 - a.) Connect the GREEN/YELLOW wire to terminal G in the upper right.
 - b.) Connect the BLUE wire to terminal N in the upper left.

- c.) Connect the RED wire to terminal L1 in the lower left.
- d.) Connect the BLACK wire to terminal L2 in the lower right.
- 2.) Replace the upper access panel previously removed and fasten it with the screws saved above.

UPS (REPO) INTERLOCK

Locate the REPO Cable (2139060) supplied with the kit. Connect UPS Interlock circuit as follows:

- Install the mating REPO connector supplied onto the end of the cable pre-terminated with mating pins.
- Plug the cable into the REPO connector at the top right (rear view) of the UPS.

Section 3.0 Mounting UPS Disconnect Panel on Liebert UPS

Locate the UPS Disconnect Panel in the kit and mount it as follows (Refer to Sheet 2 of 2142192IDW, on [page 83](#)):

- 1.) Attach mounting bracket (P/N 2142365) to disconnect panel as shown.
- 2.) Remove the lower right side knockout from the UPS Disconnect Panel.
- 3.) Locate the 5 Phillips head screws that hold the top of the lower half panel to the rear of the UPS.
They are horizontally aligned just below the fan ports. The center screw is a slightly higher than the 2 outer ones on either side.
- 4.) Remove the center one and the one immediately to its left and save them.
- 5.) Hold the UPS Disconnect Panel up to the rear of the UPS between the Battery and Output circuit breakers.
- 6.) Find the 2 holes in the top of the UPS Disconnect Panel mounting flange that align with the 2 holes of the screws removed immediately above.
- 7.) Use the 2 screws saved above to mount the panel to the UPS. Install nipple through lower right opening to fasten box to 90° elbow as shown.

Section 4.0 UPS Disconnect Panel Connections

Make the following connections in the UPS Disconnect Panel as described below. Refer to Sheet 1 of 2142192IDW, on [page 82](#).

4.1 UPS Disconnect Panel Wiring

Note: Steps 1 through 4 cable are from UPS TB1-6,7,8,9. Steps 5 and 6 are REPO wires.

- 1.) Connect the GREEN/YELLOW (#10) wire to terminals GND.
- 2.) Connect the BLUE (#10) wire to terminal NEUT.
- 3.) Connect the BLACK (#10) wire to lower connection in CB D30.
- 4.) Connect the RED (#10) wire to upper connection in CD D30.
- 5.) Connect the BLACK (#16) wire from the REPO to 21NC on the top of M2 contactor (32 NO). Reference [Figure 2-5 on page 28](#).

- 6.) Connect the RED (#16) wire from the REPO to 22NC on the top of M2 contactor (31 NO).
Reference [Figure 2-5 on page 28](#).

Note: If the REPO circuit is wired to the wrong contacts in the Disconnect Panel, the power at the Main A1 panel won't latch.

4.2 UPS Disconnect Panel Control Cable to UPS Start/Stop Control Panel

Pull control cable (P/N 2139059 coming from control panel) through 1 disconnect panel and connect to UPS Start/Stop panel as follows:

- 1.) Connect the GREEN/YELLOW (#16) wire to terminal GND.
- 2.) Connect the BLACK (#16) wire to terminal F1.
- 3.) Connect the RED (#16) wire to the M2 Coil terminal.
- 4.) Connect the BLUE (#16) wire to the Neutral terminal.
- 5.) Tighten strain relief onto cable.

4.3 UPS Disconnect Output Cable to PDU

Pull UPS Output cable (P/N 2139056) going to the PDU through the 3/4" conduit in bottom of UPS disconnect panel and connect as follows:

- 1.) Connect the GREEN/YELLOW (#10) wire to GND.
- 2.) Connect the BLUE (#10) wire to neutral (NEUT).
- 3.) Connect the RED (#10) wire to terminal 4T2 on the M2 contractor.
- 4.) Connect the BLACK (#10) wire to terminal 2T1 on the M2 contractor.
- 5.) Tighten strain relief onto cable.

Proceed to the appropriate PDU Chapter for the next steps of installation.

The A1 to UPS assembly, Interlock Cable (P/N 2142579) and Control Cable (P/N 2139059) will be installed in [Chapter 8 - A1 Disconnect Panels](#).

Chapter 5

Global PDU (GPDU - 2113764)

Perform the following connections at the GPDU. Refer to Sheet 3 of 2142192IDW, on [page 84](#).

- 1.) Turn the 480V main power OFF. Use Tag and Lockout procedures.
- 2.) Remove the front cover of the PDU. Remove the front panel of the power distribution panel in the lower front left of the PDU.
- 3.) Remove the relay control PWB from the back of the secondary box by removing connectors J1, J2 and J3.
- 4.) Remove and save the four screws, which mount the relay control board mounting bracket to the back of the secondary box.

The relay control board assembly may now be rotated out of the way to gain access to the UPS terminal block on the back wall of the secondary box.



CAUTION

Use care not to damage leads going to the transformer on the left leads side of the relay control board assembly).

- 5.) Remove the 2 position terminal block located under the Relay Control Board and replace it with the 2 position, compression type block supplied in the kit.
- 6.) Locate the 5 circuit breakers in the top of the power distribution panel. Detach the wires at the top terminal of CB1 and the top left terminal of CB4.
- 7.) Cut the cable ties in the harness going back to the panel input filter.
 - a.) Do NOT disturb the connections at the filter end.
 - b.) Verify that the above 2 wires are both on the same phase by virtue of coming from the same terminals of the input filter.
 - c.) Route these wires down to the new UPS terminal block and fasten them both to the top terminal marked L1. (This terminal is L1 on the PDU end and L2 at the UPS end.)
- 8.) Detach the wires at the top right terminal of CB4 and the top of CB5.
 - a.) Verify that the above 2 wires are both on the same phase by virtue of coming from the same terminals of the input filter.
 - b.) Route these 2 wires down to the new terminal block and fasten them both into the top terminal marked "L3". (This terminal is L3 at the PDU end and is L1 at the UPS end.)
- 9.) Remove the two access plates from the bottom of the secondary box used for cable access.
 - a.) Locate the two new replacement plates in the kit. (P/N 2142725)
 - b.) Locate and install one conduit fitting in each plate. Use existing hardware to install plates.
- 10.) Pull the "UPS Input Cable" into the PDU and up into the bottom of the left power distribution panel, through the left access plate and tighten cable connector.
 - a.) Connect the BLACK wire to the bottom of terminal of L1 and the RED wire to the bottom terminal of L3.
 - b.) Locate the 2 large dual compression type terminals at the center of the ground bus on the right side of the panel.
 - c.) Connect the GREEN/YELLOW wire to one location on the rear terminal.
 - d.) Connect the BLUE wire to one location on the front terminal.

- 11.) Pull the "UPS Output Cable" into the PDU and up into the bottom of the panel through the right access plate and tighten cable connector.
 - a.) Connect the GREEN/YELLOW wire to the unused location in the rear terminal of the ground bus.
 - b.) Connect the BLUE wire to the unused location in the front terminal of the ground bus.
 - c.) The RED and BLACK wires each have 2 wires coming from them.
 - 1.) Connect one RED wire to the top terminal of CB1.
 - 2.) Connect the other RED wire to the top left terminal of CB4.
 - 3.) Connect one BLACK wire to the top right terminal of CB4.
 - 4.) Connect the other BLACK wire to the top terminal of CB5.
- 12.) Locate the wires going from the lamps L1 & L3 in the Power Distribution Panel in the PDU to terminals L1 & L3 on the panel input filter.
 - a.) Cut only these 2 wires near the filter end.
 - b.) Tape the ends of the wires that remain on the filter.
 - c.) Route the 2 wires coming from the lamps over the tops of the breakers.
 - 1.) Connect the L1 wire into the top terminal of CB1.
 - 2.) Connect the L3 wire into the top right terminal of CB4.



Figure 5-1 Global Power Distribution Unit (GPDU)

- 13.) Next, follow this manual to the proper A1 panel connection procedure.

Chapter 6

Compact PDU (CPDU - 2133533 or 2269902)

Perform the following connections at the CPDU. Refer to Sheet 7 of 2142192IDW, on [page 88](#).

- 1.) Turn the 480V main power OFF. Use Tag and Lockout procedures.
- 2.) Remove the front cover of the PDU.
- 3.) Remove the Jumper wires in the A3 TB2 UPS (positions 1-3, 2-4).
- 4.) Pull the UPS Input Power Cable P/N 2139051 into the PDU and up into the bottom of A3 TB2 UPS, through the outer ring in the A4 access plate and tighten cable connector.
 - a.) Connect the RED wire to the bottom of terminal in TB2-1 and the BLACK wire to the bottom terminal in TB2-2.
 - b.) Connect the GREEN/YELLOW wire to one location on the GND terminal.
 - c.) Connect the BLUE wire to one location on the NEUTRAL terminal.
- 5.) Route the UPS Output Power Cable (2139056) into the PDU, through the inner cable clamp in the A4 panel, and up into the bottom of A3 TB2.
 - a.) Cut excess lead length from the Output Power cable as shown on sheet 7 of 2142192IDW, on [page 88](#).
 - b.) Connect the RED wire to the bottom of terminal in TB2-3
 - c.) Connect the BLACK wire to the bottom terminal in TB2-4
 - d.) Connect the BLUE wire to an unused location on the NEUTRAL terminal bus.
 - e.) Connect the GREEN/YELLOW wire to an unused location on the GND terminal bus.
- 6.) Next, follow this manual to the proper A1 panel connection procedure.

Chapter 7

NGPDU for LightSpeed (2326492-2)

Make the following connections at the NGPDU. Refer to Sheet 7 of 2142192SCH, on [page 78](#).

- 1.) Turn the 480V main power OFF. Use Tag and Lockout procedures.
- 2.) Remove the front cover of the PDU.
- 3.) Remove the jumper wires in the TS4 UPS (positions 1-6, 3-8).
- 4.) Pull the UPS Input Power Cable P/N 2139051 into the PDU and up into the bottom of PDU TS4, through the outer ring in the PDU's lower access plate (marked "Pwr to UPS") and tighten the cable connector.
 - a.) Connect the RED wire to the bottom of terminal in TS4-1 and the BLACK wire to the bottom terminal in TS4-3.
 - b.) Connect the GREEN/YELLOW wire to TS4-5.
 - c.) Connect the BLUE wire to TS4-4.
- 5.) Route the UPS Output Power Cable (2139056) into the PDU, through the inner cable clamp in the PDU's lower panel, and up into the bottom of TS4.
 - a.) Cut excess lead length from the Output Power cable and insulate similar to that shown on sheet 7 of 2142192IDW.
 - b.) Relabel the Output Power cable leads as shown.
 - c.) Connect the RED wire to the bottom of terminal in TS4-6
 - d.) Connect the BLACK wire to the bottom terminal in TS4-8
 - e.) Connect the BLUE wire to the bottom terminal in TS4-9.
 - f.) Connect the GREEN/YELLOW wire to the bottom terminal in TS4-10.
- 6.) Tighten cable access hole connectors on lower PDU panels.
- 7.) Next, follow this manual to the proper A1 panel connection procedure.

Chapter 8

A1 Disconnect Panels

Section 1.0

A1 Disconnect Panel Changes and Connections

Do the following changes and connections at the A1 Disconnect Panel and in the UPS Control Panel. Refer to Sheets 4 and 5 of 2142192IDW, on [page 85](#) and [page 86](#).

Section 2.0

Main Power Disconnection



WARNING

PULL THE MANUAL HANDLE DOWN ON A1 PANEL DISCONNECT PANEL TO DISCONNECT 480V MAINS FROM SYSTEM. USE PROPER LOCKOUT/TAGOUT PROCEDURES. THE FOLLOWING PROCEDURES ARE ACCOMPLISHED INSIDE THE A1 DISCONNECT PANEL.



DANGER

480V IS STILL PRESENT INSIDE BOX WITH MANUAL SWITCH PULLED. FIND THE FEEDER PANEL TO THIS ONE AND SHUT MAIN POWER OFF THERE. USE PROPER LOCKOUT/TAGOUT PROCEDURES. TEST THE A1 DISCONNECT TO VERIFY THAT ALL POWER IS OFF.

2.1 A1 to UPS Assembly Cabling for All Configurations

Locate the following cable and loose wires and pull them through the conduits as specified.

- 1.) Install a conduit fitting in the A1 Panel for the Control Cable.
- 2.) Pull the Control Cable (P/N 2139059) from the UPS Start/Stop Control Panel, into the A1 Disconnect Panel.
- 3.) Pull the Interlock Wires (P/N 2142579) from the UPS Control Panel to the A1 Disconnect Panel.

2.2 Interlock Wire Connections in the UPS Control Panel

Make the following Interlock wire connections in the UPS Start/Stop Control Panel. Use Lead Data P/N 2142579:

- 1.) Connect the GREEN/YELLOW wire to the GND terminal.
- 2.) Connect the BLACK wire to the 24NO terminal.
- 3.) Connect the PURPLE wire to the 23NO terminal.
- 4.) Connect the RED wire to the COIL terminal.
- 5.) Connect the YELLOW wire to the SEO (System Emergency Off) terminal.

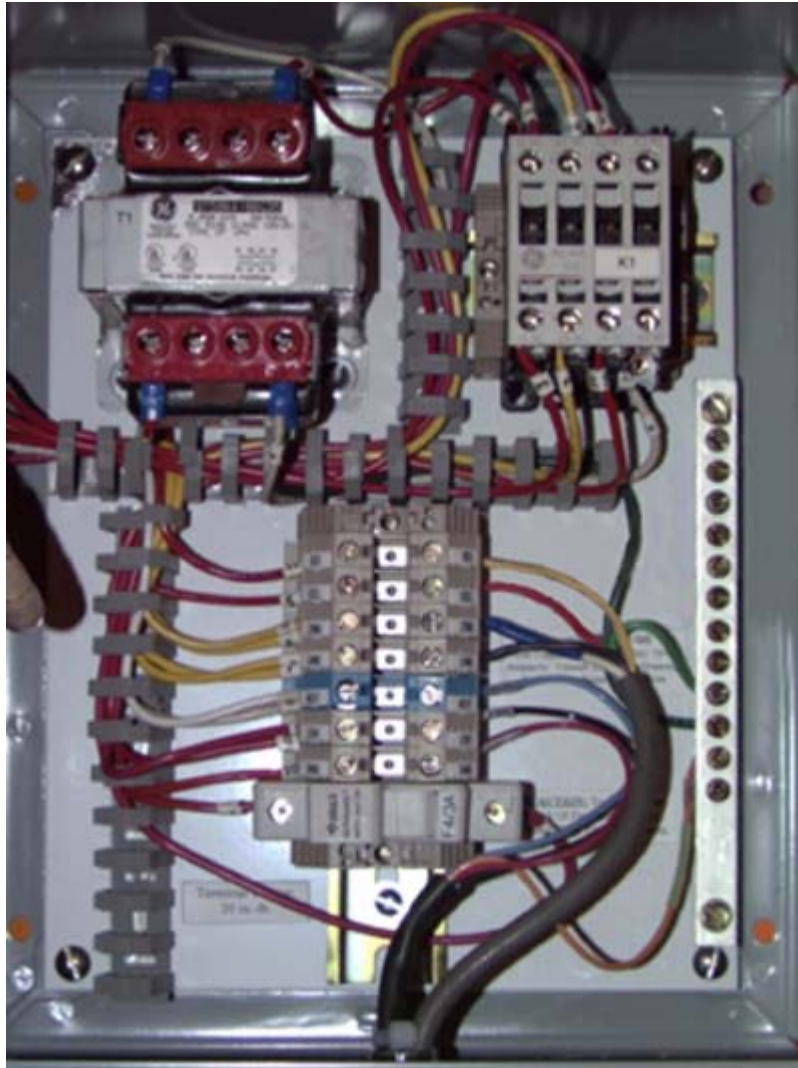


Figure 8-1 UPS Start/Stop Disconnect Panel

2.3 Control Cable Connections in the UPS Control Panel

Make the following Control Cable connections in the UPS Control Panel. (P/N 2139059).

- 1.) Connect the RED wire to the 34NO terminal.
- 2.) Connect the BLACK wire to the 120V terminal.
- 3.) Connect the BLUE wire to the neutral (N) terminal.
- 4.) Connect the GREEN/YELLOW wire to the GND terminal.

Section 3.0

Model R4502AD A1 Disconnect (GE# 308E112EAFA)

Modify the control circuit wiring and make connections in the A1 Disconnect Panel as follows:

- 1.) Do not disturb the following;
 - a.) The common side wiring going from the power indicator lamp of the main contactor coil to the common of the control transformer secondary to the cabinet GND block (RED wire together with the WHITE wire).
 - b.) The RED wire going from the other side of the power indicator lamp to the other side of the main contactor coil (high side of the contactor coil).
- 2.) At the main contactor Auxiliary Switch, disconnect the BLACK/RED wire coming from the START switch. Wire nut the free end of the RED wire. (It will not be used in the new circuit.)
- 3.) Disconnect the RED wire going from the high side of the main contactor coil to the main contactor auxiliary contact at both ends. (Discard this wire with ring terminals)
- 4.) Disconnect the BLACK/WHITE wire coming from the START switch at the AUX contact on the main contactor. (Wire nut this wire to the RED wire coming from the UPS Control Panel.)
- 5.) The BLACK wire coming from the STOP switch should be going to the other side of the room SEO's Terminal Block (SEO Jumper), next to the RED wire coming from the Control Transformer fused output. Leave it in place.
- 6.) Connect the PURPLE wire from the UPS Control Panel to the high side of the contactor coil.
- 7.) Disconnect the room SEO jumper wire from the Control Transformer fused output placed at the room SEO's Terminal Block (disconnect only this end of the jumper).
- 8.) Connect the BLACK wire from the UPS Control Panel to the Control Transformer fused output at the SEO's Terminal Block (to the RED wire from the Control Transformer fused output).
- 9.) Locate the auxiliary switch in the KIT and mount it on the manual handle in the A1 Disconnect Panel with the provided hardware per publication GEH-5075 included with the switch.
- 10.) Connect the wire coming from the room SEO's Terminal Block to one side of the Terminal block provided with the KIT (disconnected in step 7 above). Connect the other side of the Terminal Board to the auxiliary switch N.O. contact.
- 11.) Connect the other side of the N.O. contact to another position of the Terminal Board. Connect the other side of the Terminal Board to the YELLOW wire in the Interlock Wiring.
- 12.) Connect the GREEN/YELLOW ground bonding wire in the Interlock Wiring to the A1 Disconnect Panel ground block.
- 13.) Locate the "CAUTION" label in the kit and apply it to the front of the A1 Disconnect Panel.
- 14.) Locate the "A1 LEVER SWITCH" label in the kit and apply it to the front of the A1 Disconnect Panel next to the Lever Switch Handle.
- 15.) Locate the "START DISABLE" label in the kit and apply it to the front of the A1 Disconnect Panel.

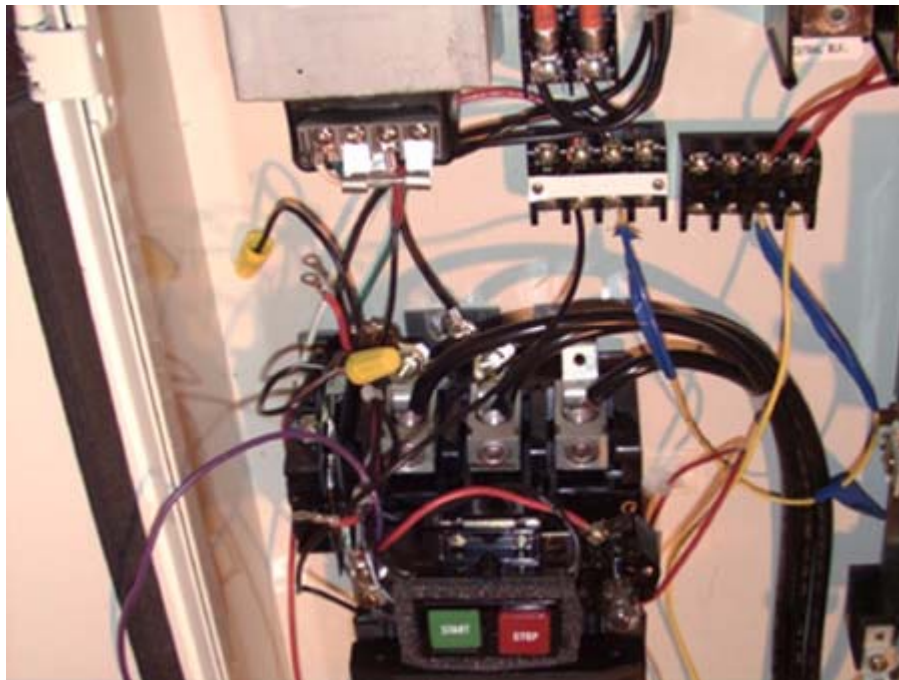


Figure 8-2 A1 Main Contactor modified for UPS



Figure 8-3 UPS A1 Manual Handle Interlock Switch

Section 4.0

HSA 125A A1 Disconnect (GE#CR308F104DCRB)



WARNING THESE MODIFICATIONS APPLY TO THE 125A DISCONNECT ONLY.

Modify the control circuit wiring and make connections in the HSA 125A A1 Disconnect Panel as follows (refer to Sheets 4 and 5 of 2142192IDW, on [page 85](#) and [page 86](#):

- 1.) Locate the MAIN CONTACTOR in the lower left quadrant of the panel.
 - a.) On the N.O. auxiliary contact at the left of the contactor disconnect both RED wires on terminal 3.
 - b.) Wire nut the free end of the RED one going back to the ON switch, It will not be used in the new circuit.
 - c.) The other wire goes to TB1. Disconnect and remove it from the harness. Discard this wire.
- 2.) Disconnect the WHITE wire on terminal 2 of the above N.O. auxiliary contact. (This wire comes from the left side of the OFF switch. It will be connected to one side of the manual auxiliary switch later.)
- 3.) Locate the 24V Control Transformer in the upper left quadrant of the panel.
 - a.) Disconnect the RED wire at X1.
 - b.) Connect the BLACK wire coming from the UPS Control Panel to X1.
 - c.) The RED wire comes from the left side of the OFF switch. Wire nut this RED wire to the RED wire coming from the UPS Control Panel.
- 4.) Disconnect the wire at TB6 going to the left coil terminal on K1 at both ends and remove it from the harness. Discard this wire.
- 5.) Connect the PURPLE wire from the UPS Control Panel to the upper left coil terminal on the "K1" Relay.
- 6.) Locate the auxiliary switch in the kit and mount it on the handle in the A1 Disconnect Panel with the provided hardware per publication GEN-5075 included with the switch. Omit step 2. The fusible switch does not have to be removed to install the interlock kit in this application.



WARNING THE FOLLOWING STEP IS VERY IMPORTANT. YOU MUST VERIFY THE WIRING CONFIGURATIONS. NO GUESSING ALLOWED.

- 7.) Examine the wiring on the terminal block TB. It has ability to install 5 different external interlock switches between terminal 1 through 6.
 - a.) The Exam Room SEO is probably between terminals 5 & 6.
 - b.) If there is a Control Room SEO it is, probably between terminals 4 & 5. (or vice versa).
 - c.) There could be other SEO's going all the way down to terminal 1.
 - d.) Determine the lowest terminal number in the series string and connect the YELLOW wire from the UPS Control Panel to it. **This needs to be very specific. No guess work.**
- 8.) Wire nut one wire from the manual auxiliary switch installed in step 6, to the WHITE wire coming from the OFF switch disconnected in step 2 above.
- 9.) Connect the other wire from the manual auxiliary switch to TB6.
- 10.) Connect the GRN/YEL ground bonding wire coming from the UPS Control Panel to the A1 Disconnect ground block.
- 11.) Locate the "CAUTION" label in the kit and apply it to the front of the A1 disconnect Panel.
- 12.) Locate the "A1 LEVER SWITCH" label in the kit and apply it to the front of the A1 Disconnect Panel next to the Lever Switch Handle.

- 13.) Locate the "START DISABLE" label in the kit and apply it to the front of the A1 Disconnect Panel.



Figure 8-4 Figure 7-4 HSA 125 Amp A1 Disconnect Panel unmodified

Section 5.0

Euro A1 Disconnect CT LightSpeed Model FCA190-A



WARNING THESE MODIFICATIONS APPLY TO THE MODEL FCA190-A DISCONNECT ONLY.

Modify the control circuit wiring and make connections in the Euro A1 Disconnect Panel as follows (refer to Sheets 4 and 5 of 2142192IDW, on [page 85](#) and [page 86](#)):

5.1 Interlock and Control Wiring from the UPS Start/Stop Panel

Reference [Figure 8-6 on page 52](#) for the following.

- 1.) Route the interlock and Control wiring into the bottom of the A1 disconnect panel using the correct electrical hardware.
- 2.) Connect the GREEN/YELLOW wire to the GND terminal.
- 3.) Connect the BLACK wire to BNC 22 terminal.
- 4.) Connect the PURPLE wire to BNC 23 terminal.
- 5.) Connect the YELLOW wire to BNC 24 terminal.
- 6.) Connect the RED wire to BNC 25 terminal.

5.2 A1 Disconnect Internal Wiring Modifications

- 1.) The A1 panel can support up to 3 SEO's. Jumpers need to be installed in each unused SEO circuit for proper operation. Reference [Figure 8-6 on page 52](#).
 - a.) Install jumper between BNC 3 and BNC 4.
 - b.) Install jumper between BNC 5 and BNC 6.
- 2.) The BNC terminal strip must be re-wired as follows. This configuration eliminates the need to cut wires allowing for easy conversion back to the original state if desired. Make sure you do not remove the wire tag identifiers for any reason. Reference [Figure 8-6 on page 52](#).
 - a.) Remove 01-BNC03 and crimped [03-BNC01, 03-BNC05] and crimped [05-BNC03, 05BNC22] and 22-BNC05. (Wires not used).
 - b.) Remove 02-BNC04 and crimped [04-BNC02, 04-BNC06] and crimped [06-BNC04, 06BNC23] and 23-BNC06. (Wires not used).
 - c.) Disconnect and tape crimped wires BNC [8-MA3, 8-R143]. (Wires not used.)
 - d.) Move wire BNC 9-R1A1 to BNC terminal 22.
 - e.) Move crimped wire BNC [1-RHA1, 1-R133] to BNC terminal 05.
 - f.) Move wire BNC 24-BNCA2 to BNC terminal 8.
 - g.) Install a jumper between BNC terminals 1 and 24. (Field Supplied)
 - h.) Install second NC Contact to the AU switch located on the PDB door.
 - I.) Install wire from AU NC-1 to BNC terminal 06.
 - II.) Install wire from AU NC-2 to BNC terminal 23.

If AU switch does not exist then install wire from BNC-06 to BNC-23
- 3.) At BNC A ensure wiring is connected as follows. Reference [Figure 8-7 on page 53](#).
 - a.) Wire 1-BNC25 to terminal 1.
 - b.) Wire 2-BNC24 to terminal 3.

This provides the normally OPEN contacts for EPO circuit of the Main Contactor.

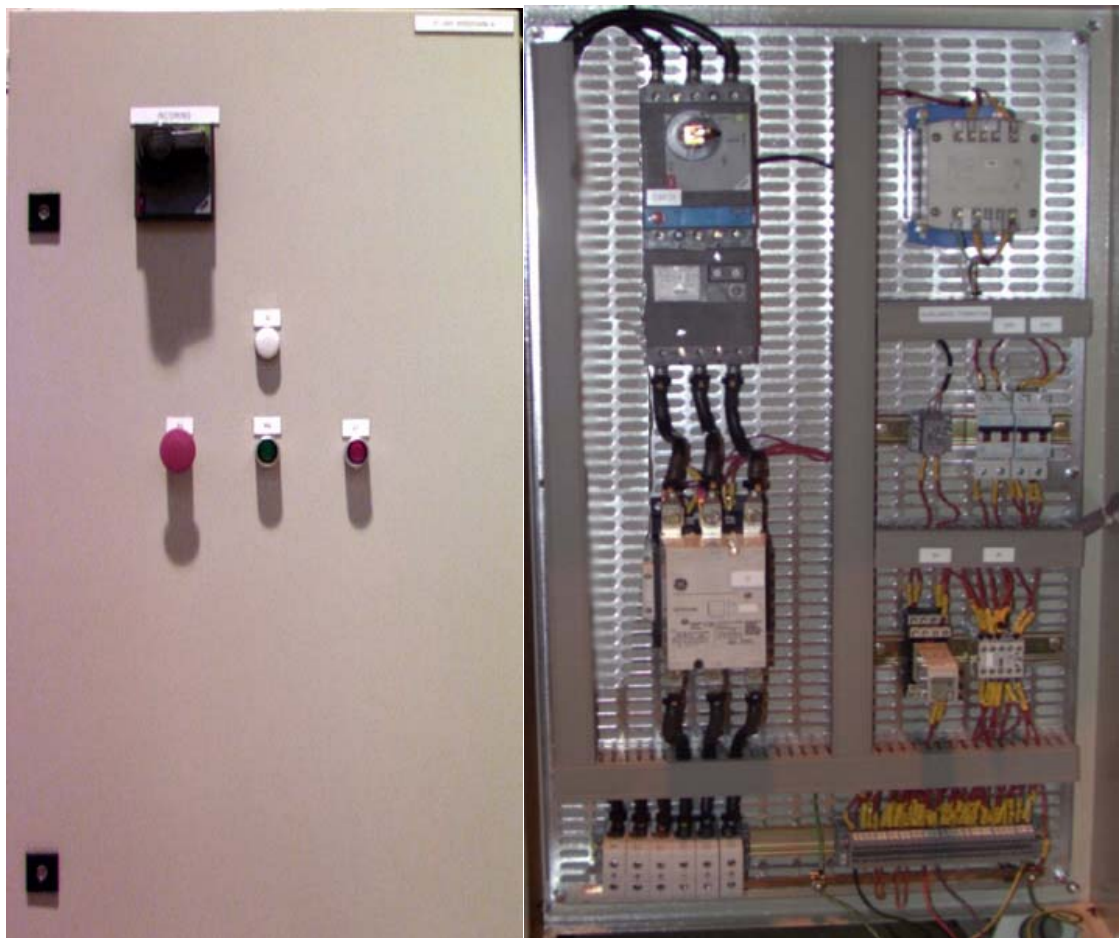


Figure 8-5 Model FCA190-A A1 Disconnect Panel Face and Interior

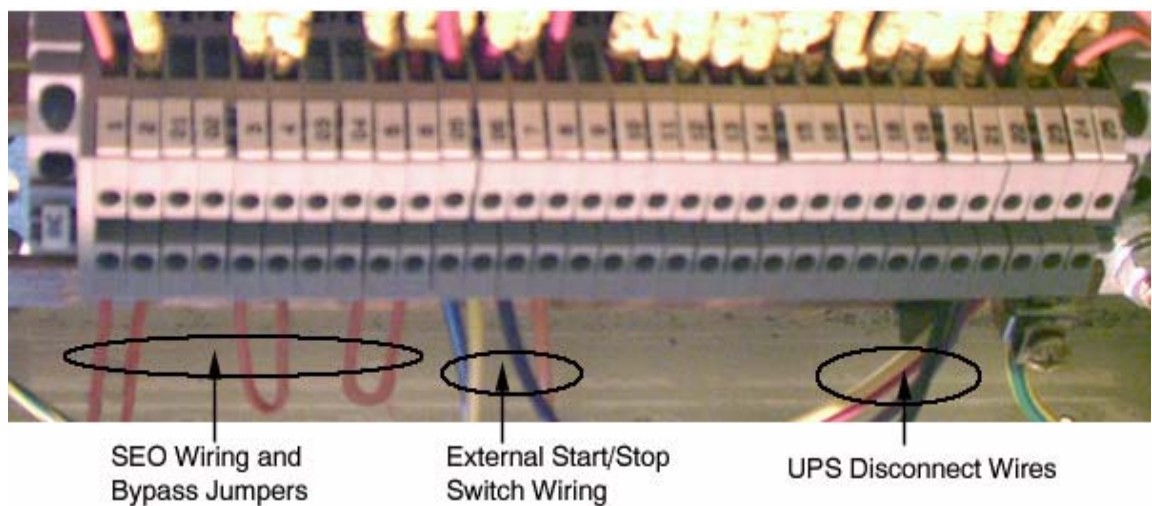


Figure 8-6 Model FCA190-A BNC Terminal Strip

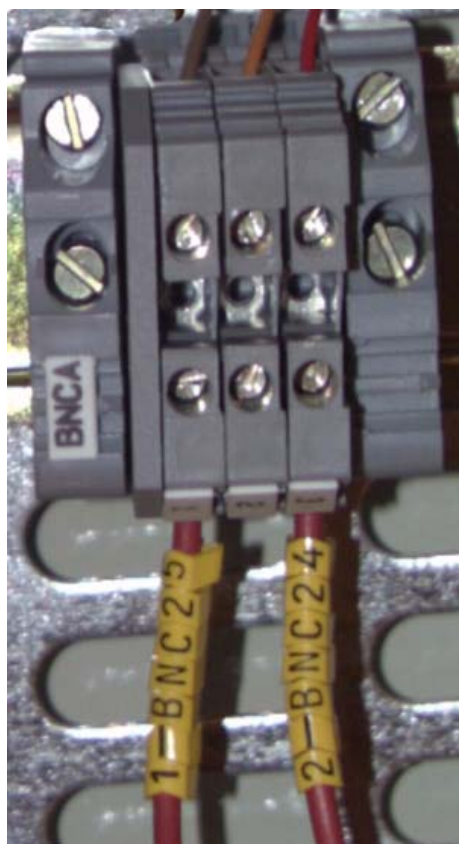


Figure 8-7 Model FCA190-A BNCA Terminal Strip

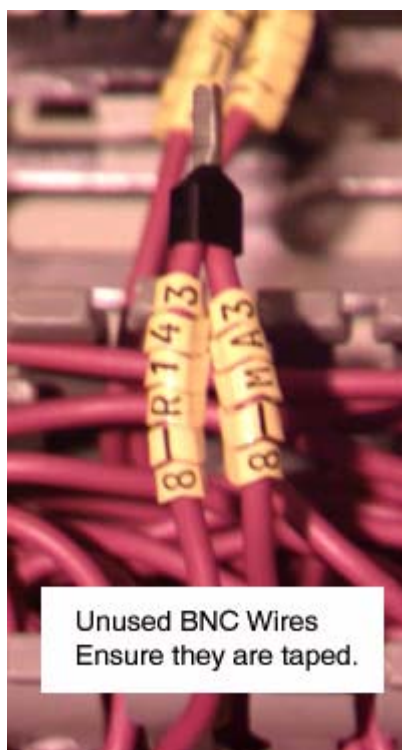


Figure 8-8 Model FCA190-A BNC Unused Wiring

Section 6.0

Euro A1 Disconnect (Model FCA190-A Predecessor)

This section is for reference only. The Coffret PDB shown is the predecessor to the Model FCA190A PDB. Electrically it is similar. This older model PDB does not have the front panel buttons. Also the wiring labels are different than the FCA190-A model. You can use the instructions found in [Section Euro A1 Disconnect CT LightSpeed Model FCA190-A on page 51](#) for modification of this style PDB.



CAUTION

When using the Section 5.0 - [Euro A1 Disconnect CT LightSpeed Model FCA190-A on page 51](#) procedure, make sure to verify your PDB wiring against the schematics provided. Failure to verify electrical wiring can result in short circuits, damaged components and possible personal injury.

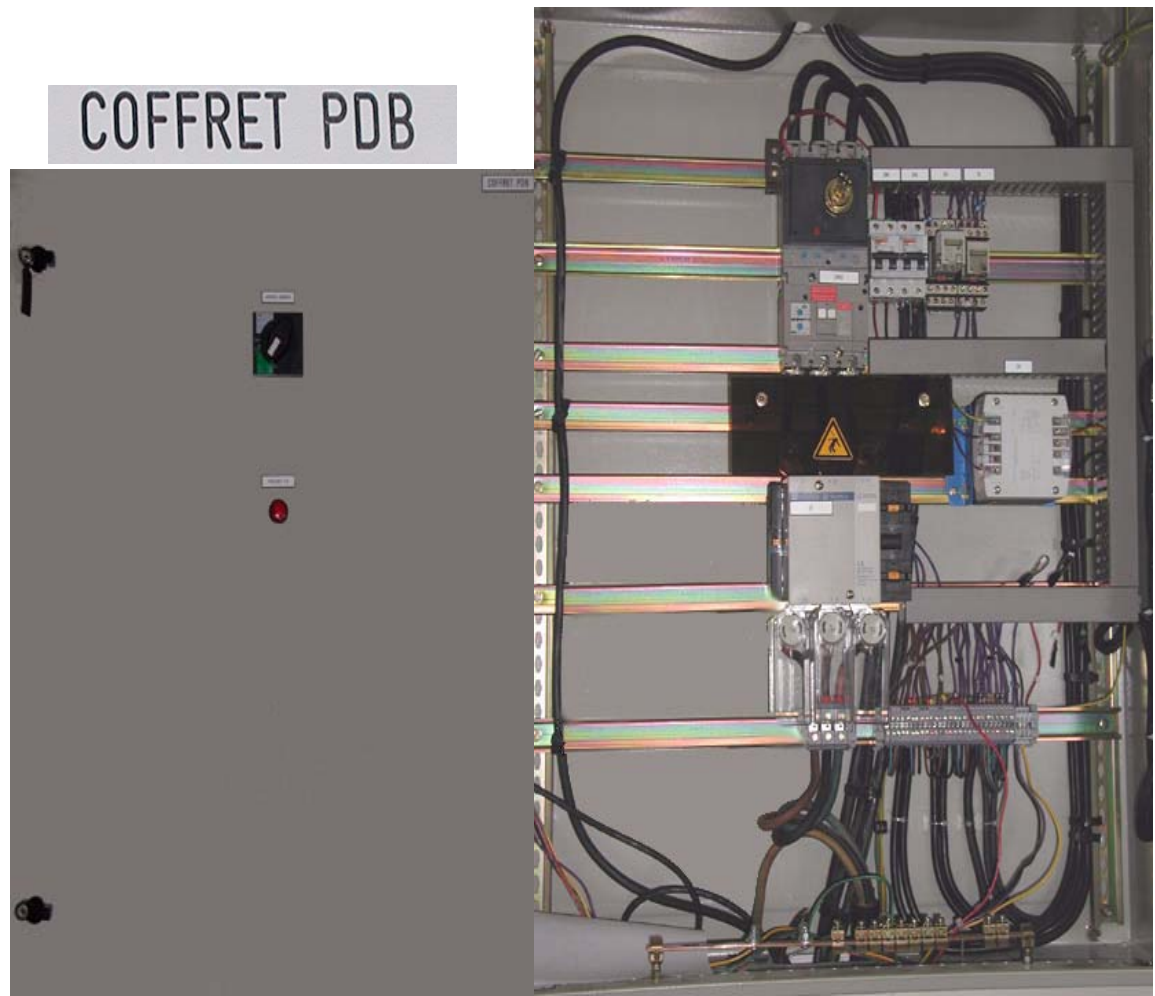


Figure 8-9 Coffret PDB Door panel and Interior layout.

Section 7.0

GE E4502AD Disconnects

7.1 Overview

Older style GE E4502AD, A1 disconnects utilized a rectangular combination “start/stop” pushbutton and a separate square neon pilot light. A present production utilizes individual round start pushbuttons, stop pushbuttons and a separate pilot light, all directly mounted to the cover. The original instructions included wiring diagrams to provide for the changing of the panel wiring to:

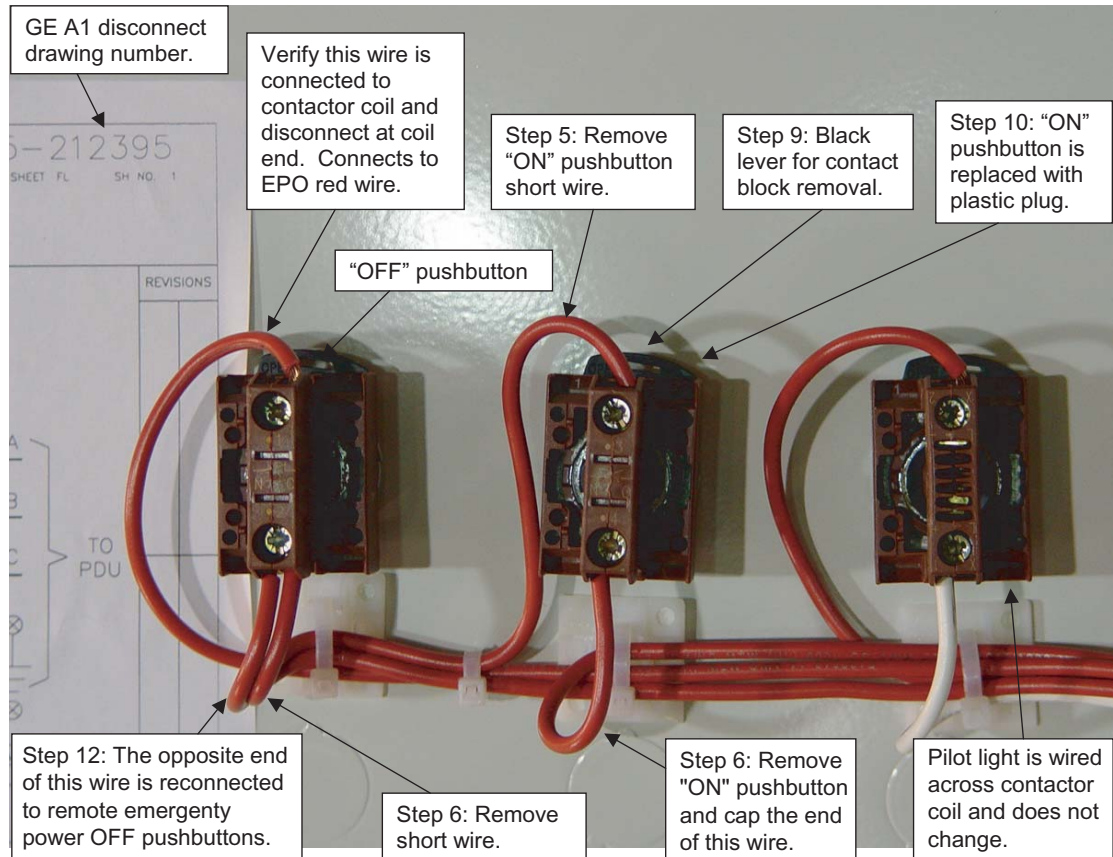
- The A1 cover mounted “EMERGENCY OFF” pushbutton into the EPO control circuit. The A1 cover mounted 2-wire “EMERGENCY OFF” pushbutton can be wired in exactly the same manner as shown on the original drawings.
- The cover-mounted pilot light two wires are already wired in exactly the same manner as shown on the original drawings. No changes are required.
- Remove the start pushbutton from the A1 cover and replace with a black plastic cover plate, catalog #P9ARHPR.

7.2 Step-by-Step Procedure

- 1.) Turn OFF power feeding the A1 panel.
- 2.) Lock OFF A1 Main disconnect handle.
- 3.) Confirm that this is a GE E4502AD panel, with wiring diagram 55-212395 mounted in the door. For other panels, the methodology of how to make this change remains the same, but the specifics would have to be determined by the installer.

REMOVE "ON" PUSHBUTTON

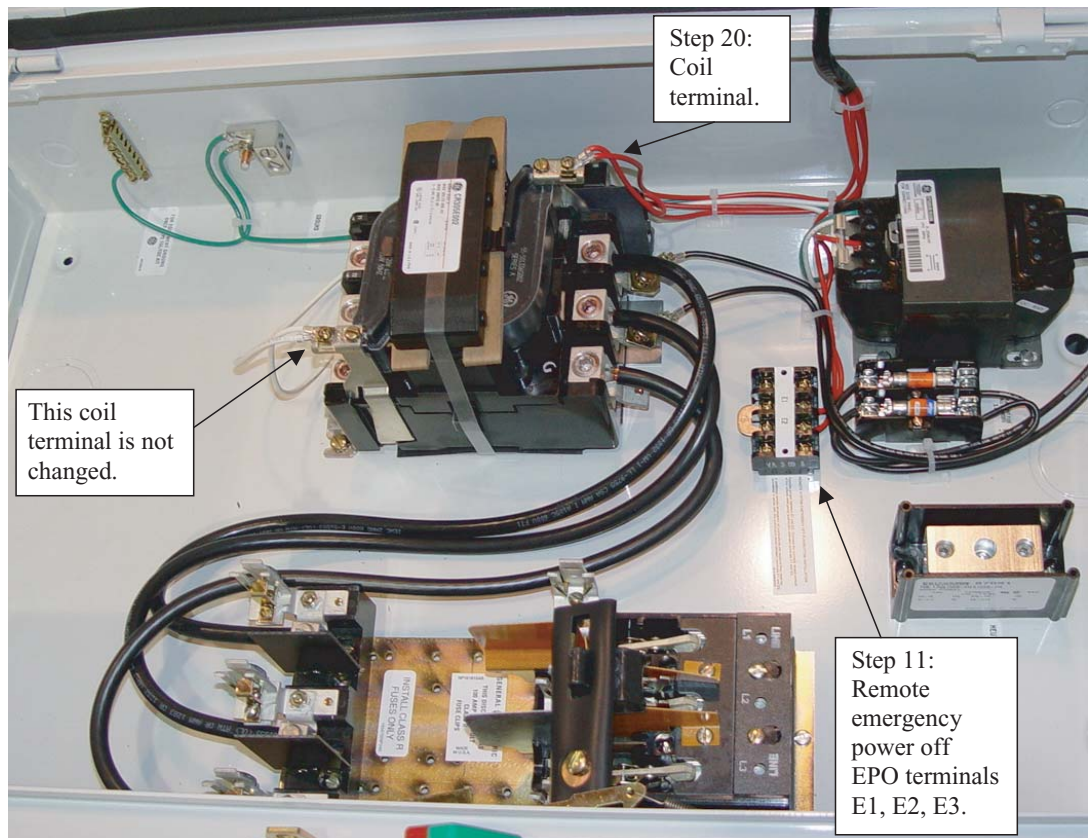
- 4.) Remove wire ties and tape applied to the wires that connect the cover-wired devices in the A1 panel, to facilitate verification of the routing of each wire.
- 5.) Locate the A1 cover mounted "ON" pushbutton and completely remove the short wire that connects one terminal from the "ON" pushbutton to the cover-mounted "OFF" pushbutton. This short wire may be discarded. Retighten the OFF pushbutton's remaining wire.



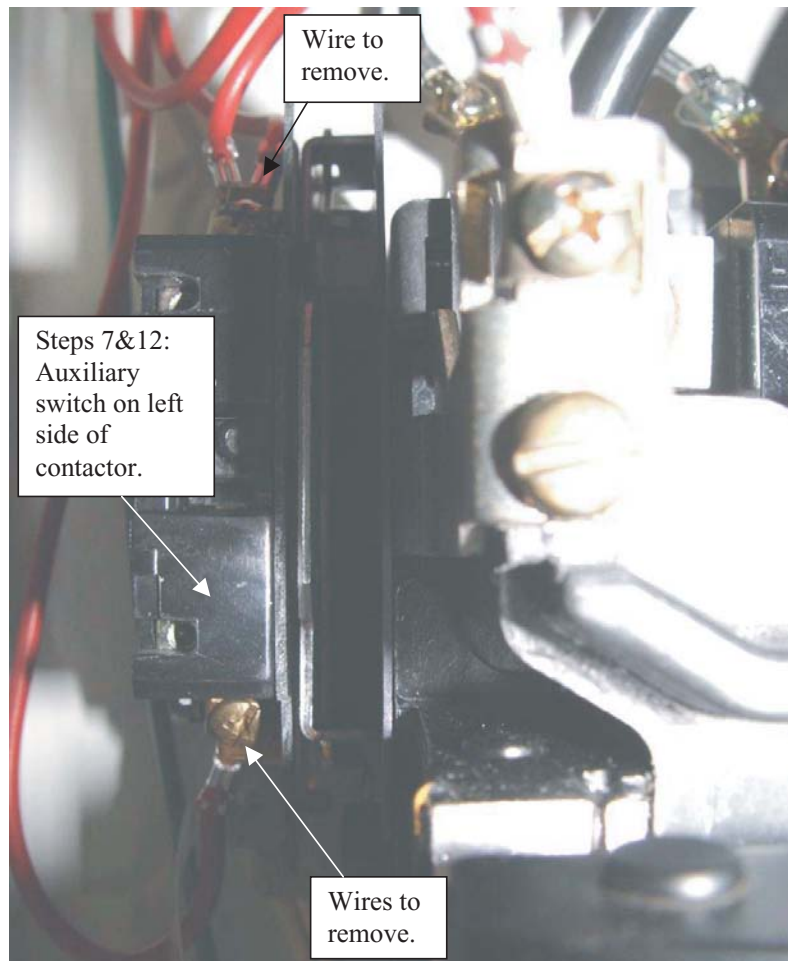
- 6.) Remove the wire from the other terminal on the A1 cover-mounted "ON" pushbutton.
- 7.) Locate the opposite end of the wire from step 6 that runs to the contactor side-mounted auxiliary contact. Wiggle the wire to confirm this is the same wire. Remove from the auxiliary contact and either wire nut both ends or remove this wire. The contactor auxiliary contact is not used in this application.
- 8.) Remove the "ON" pushbutton from the cover by locating the black lever on the brown base (just above the markings "1,2,3") and sliding the lever to the left. This exposes the locking nut, which, when unscrewed, will release the "ON" pushbutton from the cover.
- 9.) Install the Black plastic cover plate, #P9ARHPR, in the former "ON" pushbutton's hole.

WIRE "OFF" PUSHBUTTON

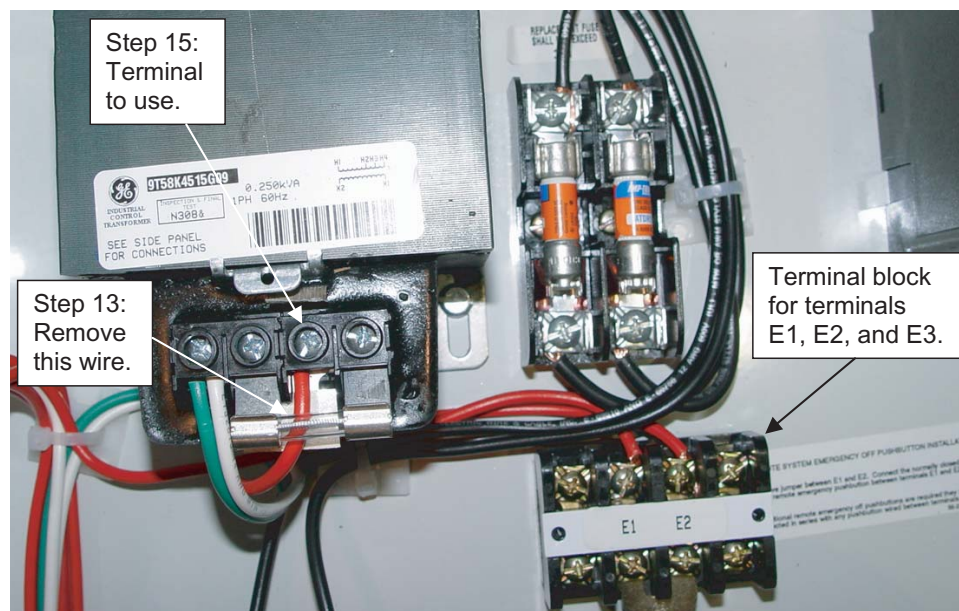
- 10.) The OFF pushbutton has two remaining wires: usually one top and one bottom. Starting with the terminal from which the short wire was removed (in step 5, above), trace the wire to the other end at the auxiliary contact, and remove this end, which is connected in the panel. Using a wire nut, connect this wire to the RED wire from the UPS control panel that is marked "COIL". The original drawing, sheet 3, shows the panel wire as: BLK/WHT designation.
- 11.) The remaining wire from the A1 cover-mounted "OFF" pushbutton should be traced and verified that it is in series with the existing remote Emergency-Off pushbuttons. This is usually at terminal block E2. If it is not connected to terminal block E2 (or if it is an earlier A1 panel without a terminal E2), verify -- and if necessary, reconnect to -- the external REMOTE Emergency-Off pushbuttons designated as EPO on sheet EPO circuits in place of the black wire (BLK) shown on sheet 3. See the illustration below.



- 12.) Remove the short red wire from the main contactor auxiliary switch at both ends and discard the wire. (See illustration, below)



- 13.) Remove the red wire from the A1 panel control transformer secondary fuse typically third from the left. See the illustration below.



- 14.) Remove the other end of the red wire removed in step 13 at the A1 panel remote EPO block terminal marked E1. Discard the wire.
- 15.) Install BLACK wire from the “UPS Control Panel” to control transformer on the secondary of the fuse referred to in step 13.
- 16.) Locate the auxiliary switch kit for the A1 disconnect and mount it on the fusible sw operator mechanism in according with instructions GEH-5075 for 100A disconnects.
- 17.) Install one wire between the disconnect mounted auxiliary switch and an unused terminal on the A1 EPO terminal block. Mark unused terminal E3 if not already marked.
- 18.) Install one wire from the A1 disconnect mounted auxiliary switch to non-wired terminal at E1 on EPO terminal block.
- 19.) Install YELLOW wire from UPS control Panel to terminal block terminal E3 on A1 disconnect EPO terminal block.
- 20.) Connect PURPLE wire from the UPS control panel to the contactor coil. This coil terminal is on the coil and is the high side terminal located at the upper left corner of the contactor. Do not connect to the terminals with the white wires.
- 21.) There are no changes to the pilot light wiring in that it is already factory wired in Parallel with the contactor coil.
- 22.) Connect the GREEN/YELLOW ground bonding wire from the UPS control panel to the A1 Disconnect Panel ground block.
- 23.) Locate the “CAUTION” label in the kit and apply it to the front of the A1 Disconnect Panel.
- 24.) Locate the “A1 LEVER SWITCH” label in the kit and apply it to the front of the A1 Disconnect Panel next to the Lever Switch Handle.

7.3 Check Out

- Tape the wires going to the A1 panel cover OFF pushbutton and pilot light in the vicinity of the door hinge so as to avoid abrasions.
- With continuity tester, confirm loss of continuity when the EPO is pressed between EPO and coil, as shown on 2142192SCH, sheet 3.
- Visually confirm that there are no wires that have not been capped off in the panel, or on the inside of the cover.

Section 8.0

Unique System Disconnect Panels

If the system disconnect panel is different from the models described previously in this document, the following instructions are to be used to modify the existing system disconnect panel for connection to the UPS Control Panel. The following information assumes a two button (ON,OFF) control panel with a latching three pole contactor and a manual shut off handle.

MODIFICATION INSTRUCTIONS

Reference [Figure 8-10 on page 61](#) for a generic main contactor circuit. The existing circuit for the main contactor control should be similar to the one shown. The instructions described in this section refer to [Figure 8-10](#) and [Figure 8-11](#).

- 1.) Disconnect the "Hi" side of the main contactor coil power source from the "OFF" switch and connect to the UPS Control Panel terminal labeled "24 NO." This line shall have fuse protection (5 Amps Maximum) as indicated by "F1". The voltage on this line shall not exceed 300 VAC.
- 2.) Disconnect the lead going to the "Hi" side of the main contactor coil from the "ON" switch or main contactor auxiliary switch. Connect this lead to the UPS Control Panel terminal labeled "23 NO".
- 3.) Connect the un-terminated side of the "OFF" switch, (one side of the OFF switch was left un-terminated as a result of step 1 to the UPS Control Panel terminal labeled "COIL".
- 4.) Install a position sensing switch on the manual shut off handle of the existing system disconnect panel. The switch shall be a momentary, normally open, single pole switch. The switch shall be installed in such a way that the manual shut off handle will actuate this switch to the normally closed position when the manual shut off handle is in the "ON" position. The switch shall return to the normally open position when the manual shut off handle is in the "OFF" position. This switch shall have a minimum rating of 5 AMP and 100 VAC. The manual handle position sensing switch shall be wired in series with the System Disconnect Panel "OFF" switch and the Room SEO (System Emergency Off) switches.
- 5.) Locate the incoming SEO switch leads at the existing system disconnect panel. One end of the series loop will be connected to the "OFF" switch. The other end will be connected to the "ON" switch or main contactor auxiliary switch. Disconnect the end of the lead connected to the "ON" switch or main contactor auxiliary switch.
- 6.) Connect one side of the manual handle position sensing switch to the SEO switch lead disconnected in step 5. Connect the other side of this switch to the UPS Control Panel terminal marked "SEO."

Note: In the new circuit the main contactor auxiliary switch and "ON" push button switch in the system disconnect panel are no longer used.



DANGER



FAILURE TO INSTALL THE ABOVE MENTIONED POSITION SENSING SWITCH WILL ALLOW THE UPS TO SUPPLY LETHAL POWER TO PORTIONS OF THE SYSTEM, WITH THE MANUAL SHUT OFF HANDLE IN THE "OFF" POSITION. IF IT IS NOT POSSIBLE TO INSTALL THIS SWITCH, ALL PERSONNEL USING OR SERVICING THIS SYSTEM MUST BE AWARE THAT AN SEO OR OFF SWITCH ON THE SYSTEM DISCONNECT PANEL MUST BE DEPRESSED TO DISABLE THE UPS. THIS IS VERIFIED BY OBSERVING THE "UPS ON" LAMP ON THE UPS CONTROL PANEL GO FROM ON TO OFF.

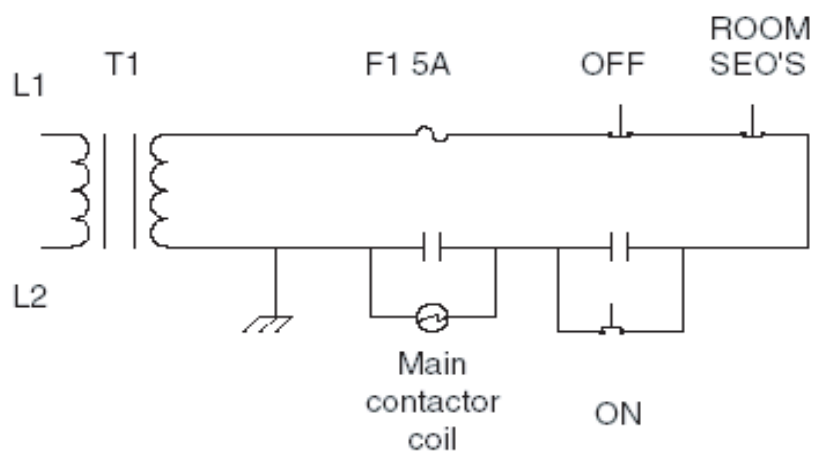


Figure 8-10 Generic Main Contactor Disconnect Unmodified

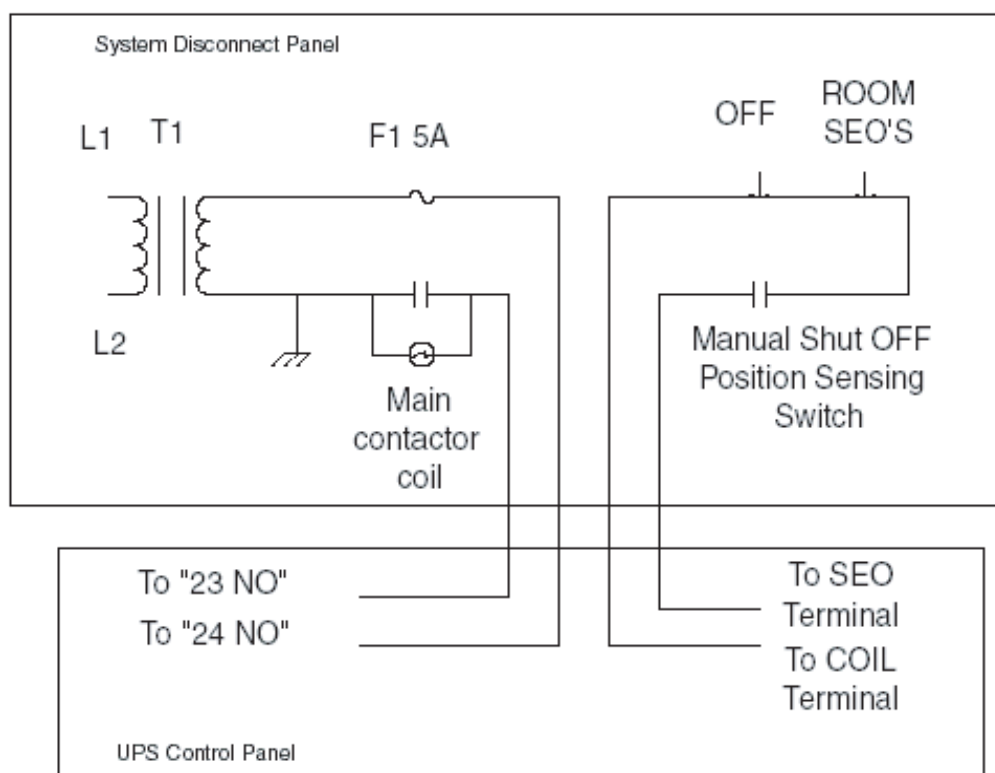


Figure 8-11 Generic Main Contactor Disconnect Modified

Section 9.0

Non-Standard Installations

There may be several configurations of system disconnect panels used with this piece of equipment. For safe and reliable operation of the UPS system, the final modification to the system disconnect panel must match the circuit shown in [Figure 8-11 on page 61](#).

Due to site requirements, it may be necessary to mount either the Disconnect Panel or Control Panel in different places from the previously specified. If Output or Input cables need to be replaced, make sure the new cables follow local electrical compliance.

Chapter 9

Start Up Procedure

Use this procedure to bring the system up the first time power is applied after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off and turn PDU Circuit Breaker (CB) OFF. This keeps the system computers from needlessly booting up until proper UPS system operation is verified. Note: It is strongly recommended to have the User's Guide for the particular UPS been used.

Check the following before proceeding to power up the UPS unit:

- 480V main power is OFF. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the procedure.
- Verify that the CT System is ready to receive power and all personnel are clear of the machine.

IMPORTANT Powerware UPS warranties require that a Powerware representative run a free “Start Up and Test” procedure. Call Powerware at 800-843-9433 to schedule an appointment. Failure to perform this checkout will void the UPS warranty.

Section 1.0

Powerware 9150

- 1.) Turn UPS “OFF” via On-Off toggle switch on its front control panel (under plastic cover)
- 2.) Remove the UPS front cover panel and turn the Maintenance Bypass switch to the “SERVICE” position.
- 3.) Turn “ON” the UPS battery circuit breaker(s), CB1 & CB2.
- 4.) Verify the CT System is ready to receive power and all personnel are clear of the machine.
- 5.) Turn “ON” utility power at the main A1 panel by setting the manual disconnect to the “ON” position.
- 6.) Press the “START” button on the UPS Control Panel. Verify the “UPS ON” light turns on and the system begins to power up.

Note: If the system fails to start up or the A1 disconnect does not stay energized, check the following:

- Verify no system SEO is locked open.
- Verify CB1 and F1 in the UPS Disconnect Panel are OK.
- Verify F4 in the UPS Control Panel is OK.
- Verify all fuses in the A1 Disconnect Panel are OK.
- Verify the PDU main control circuit fuses / circuit breaker(s) are OK.
- 7.) Start the UPS by setting the UPS On-Off toggle switch to the “ON” position.
- 8.) The UPS should begin its internal startup sequence and synchronize to the utility power line. It will beep periodically while it completes internal startup. After about 3 to 5 minutes it should display UPS ON, LINE ON, LOAD, and BYPASSED indicators.
 - If the UPS does not start and an LED other than the UPS ON indicator is blinking, the UPS is in Automatic Configuration Mode. Refer to [AUTOMATIC CONFIGURATION MODE on page 64](#) or the UPS User's Guide for more detailed information.
 - If the ALARM LED is On or Blinking (11 short beeps and 5 long beeps), there is an input phase error. The main input phases must be rotated at the PDU MAINS INPUT Terminal Block. Do NOT reverse the connections at the UPS interface.
 - Refer to the UPS User's Guide for a detailed listing of diagnostic information for the UPS.

- 9.) Turn the Maintenance Bypass Switch fully counter-clockwise to the "UPS" position (normal position).



CAUTION

Before rotating the Maintenance Bypass switch to the UPS position, verify that the UPS ON, LINE ON, LOAD, and BYPASSED indicators are lit.

- 10.) The BYPASSED indicator should turn off when the power is transferred to the inverter, after approximately 30sec.
- 11.) Reinstall the front cover of the UPS.

Note: The UPS is now providing 120VAC power to the Table, Gantry and Console. In the event of a momentary mains power loss, scanning will be aborted / inhibited. However, the UPS will continue to provide power to the Table, Gantry & Console for at least 10 minutes. The UPS will "beep" occasionally, letting you know that it is on BATTERY BACKUP. If mains power is restored prior to shutting down the CT system, the UPS and CT system will automatically return to normal operation.

AUTOMATIC CONFIGURATION MODE

Automatic configuration is performed during UPS startup. The UPS attempts to match its output voltage and frequency to the input power.

When the input voltage and frequency has been determined, the UPS beeps twice and sequences through all the LEDs. One of the LEDs remains blinking. The blinking LED shows the configuration, which has been automatically selected (see the configuration table below).

If the automatic selection is the same as the latest previously used (reference) configuration, the LED blinks for only 10 seconds. The sequencing LEDs are displayed again and the UPS starts. If the automatic selection differs from the latest previously used (reference) configuration setting, the LED blinks (showing automatic configuration selection) until the RESET pushbutton is pushed.



CAUTION

Verify the selection is correct for the CT system. It must be 120/208V, 50 or 60 Hz, depending on input frequency.

If the selection is correct, push the RESET button momentarily to store this selection in the UPS. The UPS beeps twice and sequences the LEDs. The LED indicating the selection blinks for 10 seconds and the LEDs sequence once again and the UPS starts. This automatic startup can be interrupted by pressing the RESET pushbutton quickly during the 10 seconds; the UPS changes to the Manual Configuration Mode. Manual configuration is not recommended. Refer to the UPS User's Guide for further information.

Blinking LED	Input/Output Voltage Configuration	Output Frequency
ALARM	120/208V	50 Hz
SERVICE	110/220V	50 Hz
OVERTEMP	127/220V	50 Hz
OVERLOAD	100/200V	50 Hz
80%	120/240V	50 Hz
60%	120/208V	60 Hz
40%	110/220V	60 Hz
LOAD	127/220V	60 Hz
BYPASSED	100/200V	60 Hz
ON BATTERY	120/240V	60 Hz

Table 9-1 Blinking Configuration

Section 2.0

Powerware Plus Model 12/12

This UPS should have been configured at the factory for Automatic Bypass and Maintenance Bypass. Verify or change this setting, before powering up the unit.

- 1.) On the Top of the UPS, lift the cover by grasping latch, pulling forward and upward.
- 2.) Verify that the UPS Input Circuit Breaker (CB1), Battery Breaker (CB2) and Output Breaker (CB3) are in the "OFF" state. Turn off if necessary. The CB1, CB2, and CB3 are located under the top cover of the UPS.
- 3.) Locate the Bypass Configuration Switch (S3) in the rear panel of the UPS.
- 4.) Set the Automatic Bypass and Maintenance Bypass (S3) switch to the Enable (UP) position.
- 5.) Turn "ON" utility power at the main A1 panel by pulling the manual level "UP" to the "ON" position.
- 6.) Turn "ON" the UPS Input Circuit Breaker, CB1.
- 7.) Turn "ON" the Output Breaker (CB3) and Battery Circuit Breaker (CB2).
- 8.) Press "START" on the auxiliary UPS Control Panel on top of the UPS. All indicator lights illuminate for approximately ten seconds.
- 9.) The Bypass light will flash in the event of a phase rotation error. The input phases must be rotated at the PDU Terminal Block. (See Sheet 3 of 2142192IDW for Global PDU connections and Sheet 7 of 2142192IDW for Compact PDU connections).
- 10.) Turn the UPS "ON" by depressing the "ON" button. The HORN beeps and the BATTERY & BYPASS lights flash as the Rectifier turns on.
- 11.) After approximately 15 seconds, the BYPASS & NORMAL lights will be on solid indicating that the UPS is operating in the Normal Mode.
- 12.) If the UPS does not start and an LED other than the UPS ON indicator is blinking, the UPS is in Automatic Configuration Mode.
- 13.) If the BATTERY & BYPASS lights or BYPASS & NORMAL lights flash for more than one minute or if the lights flash in any combination other than the mentioned above, contact the field service representative for the UPS unit.

Section 3.0

Liebert UPS

- 1.) Remove the back plate of the UPS to make sure the Input, Output, and Battery circuit breakers are OFF (open).
- 2.) Close the Battery circuit breaker on the rear of the UPS ("ON" position). Dynamic Bypass is standard with this unit.
- 3.) Turn "ON" utility power at the main A1 panel by pulling the manual level "UP" to the "ON" position. The load is now powered by the utility.
- 4.) Close the UPS Input circuit breaker.
- 5.) Depress the START button on the front of the UPS Control Panel and hold it in until the UPS ON lamp illuminates. This may take 5 seconds. Then release the button. The UPS starts up in the BY-PASS mode. This will supply power to the UPS control circuits. The fans will start and the LCD display will become active.
- 6.) Observe the LCD display. The UPS will prompt you to review and accept the default settings, or make selections to configure the UPS. Refer to the LIEBERT User's Manual for more details.

- 7.) After configuring, if needed, the unit will start self-tests.
- 8.) If all self-tests are 'OK', the horn will sound and the "Close Breaker then Press ON" message will appear.
- 9.) Close the Output circuit breaker (on the back of the UPS) and press the ON button.
- 10.) If all connections are correct it will automatically advance to the NORMAL mode. The "Normal Operation" message should be displayed.
- 11.) If the UPS advanced to the NORMAL mode, the PDU CB4 circuit breaker may be turned on and allow the CT System to boot up. If the UPS did not advance to the NORMAL mode and no error messages are on the UPS screen, depress the UPS ON button on the front of the UPS.
- 12.) Once the UPS is in NORMAL mode and the UPS properly assumed the load when the CB4 breaker was turned on, the installation is complete. Put the PDU front cover in place and allow the CT System to continue booting up.
- 13.) If the UPS stayed in BY-PASS and there was a message on the UPS screen indicating power phase reversal, leave the CT CB4 breaker OFF and turn the main power OFF. Tag and Lock it. At the UPS power input terminals, reverse the BLACK and RED wires. On the LIEBERT model this will require moving the UPS Disconnect Panel aside to get at the input terminal block.
- 14.) Replace the safety covers. Verify that all the UPS breakers are ON. Bring power back up as described above. The UPS should start up in BY-PASS and advance to NORMAL as described. If it does, turn the PDU CB4 breaker on. If the UPS properly assumes the load, put the PDU cover in place and allow the CT system to continue booting up. The installation is complete.
- 15.) If the UPS does not go into NORMAL or if it does not assume the CT load, the appropriate UPS Field Engineer will have to be notified. The phone numbers are located on labels on the side of the UPS or in the User's Manual.

Comment: Service for the UPS is provided by the manufacturer. GEMS does not service them. The UPS may be operated in the BY-PASS MODE until the problem is corrected. But, there is NO BATTERY BACKUP in this mode.

Section 4.0

Shut-Down Procedure

Use this procedure to bring the system up the first time power is applied after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off and turn PDU Circuit Breaker (CB) OFF. This keeps the system computers from needlessly booting up until proper UPS system operation is verified.

Note: It is strongly recommended to have the User's Guide for the particular UPS being used.

Check the following before proceeding to power up the UPS unit:

- 480V main power is OFF. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the procedure.
- Verify that the CT System is ready to receive power and all personnel are clear of the machine.

4.1 Powerware 9150

To shut down the UPS do the following:

- 1.) Remove the front panel.
- 2.) Turn the Maintenance Bypass switch to the SERVICE or BYPASS position.
- 3.) Turn "OFF" the UPS ON/OFF toggle switch (under the plastic cover).

- 4.) Turn the UPS battery circuit breaker CB1 and CB2 to the "OFF" position.

Comment: Although the UPS stops supplying power and is disconnected internally from the batteries, system power still passes through the UPS cables. Use caution when servicing.

4.2 Powerware Plus Model 12/12

If the UPS has Bypass mode disable (S3 in the OFF position), shut down the UPS doing the following:

- 1.) Hold the UPS OFF button. The HORN will sound for 3 seconds. The NORMAL light will go OFF.
- 2.) Open (OFF) the Input Circuit Breaker (CB1), Battery Breaker (CB2) and Output Breaker (CB3) to insure that all power is removed from the protected equipment.

If the UPS has Bypass mode enable (S3 in the ON position), shut down the UPS doing the following:

- 1.) Hold the UPS OFF button until the unit transfer to Bypass. The Bypass light will illuminate and the HORN will sound.
- 2.) Open (OFF) the Input Circuit Breaker (CB1), Battery Breaker (CB2) and Output Breaker (CB3) (Bypass light will go out) to insure that all power is removed from the protected equipment.

4.3 Liebert UPS

To shut down the UPS do the following:

- 1.) Press the OFF button twice to remove power to the load by turning off the inverter and opening the optional bypass.
- 2.) Note that the Output Circuit Breaker remains closed, even though output power is not available.
- 3.) When the UPS is turned off with the OFF button, the control logic and battery charging are still active. To disable the AUTO-RESTART press the ON button.

Note: If OFF button is pressed a third time, the UPS will prompt to press the OFF button a fourth time. This opens the Output Circuit Breaker at the rear of the unit.

Chapter 10

Theory of Operation

The following sequences describe the UPS Control Circuit operation.

Section 1.0

Start Sequence

- 1.) Verify that the UPS Input, Output and Battery breakers are ON. Move them to the ON position if required.
- 2.) Move manual lever on A1 Disconnect to ON, AUX switch on lever closes.
- 3.) Depress START switch on UPS Control Panel. 120V is applied to contactor M coil in A1. M pulls in and supplies 480V 3Æ power to the isolation transformer in the PDU, which in turn supplies 208/120V power to UPS input. UPS turns on in BYPASS mode. 120V is supplied to control transformer primary in UPS Control Panel. The 120V is converted to 24V and supplied to control relay K1 coil by way of the START switch contact, the enclosure cabinet door switch, AUX handle switch in A1, the room SEO's and the STOP switch in A1. This closes contact K1-1 and K1-2 in the UPS Interlock Panel (across the two halves of the START switch). Contact K1-3 at the control transformer primary also closes, interlocking the M2 coil in the UPS Disconnect Panel. This supplies 208/120V 120° power to the PDU.
- 4.) Release START switch; K1, M and M2 remain locked on because of the K1-1, K1-2 and L1-3 contact closures.
- 5.) The UPS continues through its turn-on cycle and goes into NORMAL mode.

Section 2.0

Mains Power Outage Sequence

If the 480V 3Æ main power fails or is removed upstream from the A1 Disconnect, the contactor M in A1 Disconnect Panel drops out. K1 in the UPS Control Panel and M2 in the UPS Disconnect Panel remain closed. They are powered by the UPS output, which is now powered by battery. The CT functions powered by the UPS remained live.

Section 3.0

Mains Power Return Sequence

If the 480V 3Æ power is supplied to the primary of the A1 control transformer, 120 V are applied to the contactor M coil by way of K1-2 in the UPS Control Panel. This will supply 480V 3Æ power to the PDU allowing system to return to full operational conditions.

Section 4.0

Frequency OFF Sequence

- 1.) Depress either STOP switch in the A1 Disconnect Panel, the UPS Control Panel, or any of the room SEO's, to remove power from the K1 coil. This will open the K1-1, K1-2 & K1-3 interlock contacts. M in the A2 panel will drop out removing 480V 3Æ from the PDU. The M2 AUX contact will close, which shuts down the UPS and opens its output breaker. (Also Battery Breaker on POWERWARE).
- 2.) Release the above STOP switch. The system remains OFF. A START sequence must be initiated to restart the system.

Section 5.0

Turn OFF by way of A1 Manual Lever Sequence

Pull A1 Disconnect Panel lever down. 480 3Æ is removed from the PDU. The AUX switch on this lever is also opened. An Emergency OFF sequence then also occurs, removing the 480VAC from the CT System and shutting the UPS down.

Chapter 11

Drawings

Section 1.0 - Schematic (2142192SCH, Rev 1)

2142192SCH

See the GEMS ePDM System to determine the status of this document.

NOTES:
1 THIS SCHEMATIC IS A COMPANION TO
INSTALLATION INSTRUCTION DRAWINGS
2142192GSP AND 2142192IDW

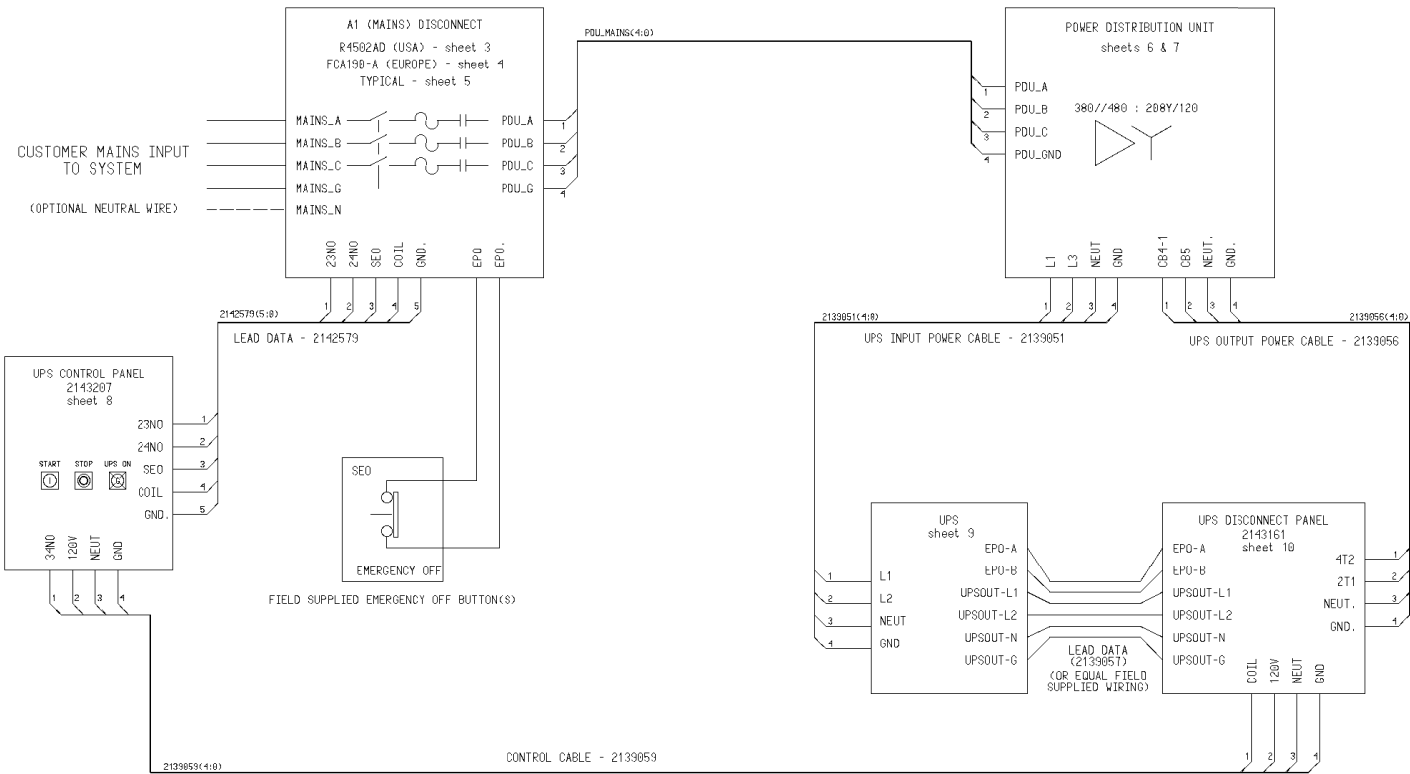
SHEET_NAME	SHT NUM	REV
UPS_IC	2	1
A1_DISC (USA)	3	1
A1_DISC (EUROPE)	4	1
A1_DISC (TYPICAL)	5	1
PDU	6	1
PDU	7	1
UPS_CNTRL	8	1
UPS	9	1
UPS_DISC	10	1

BLOCK PATHNAME \$GEMS_CT_XRAY_GEN/INTERCONNECT/2142192_SCH/UPS_IC				BLOCK SHT 1 OF 2			
REV 1	PCN 2002189	DESCRIPTION ADDED INFO FOR NGPDU-2	MADE J. R. SCHMIDT	DATE 10/DEC/2000	GE MEDICAL SYSTEMS MILWAUKEE, WI USA	DESIGN TITLE UPS INTERCONNECT	
MADE BY J. R. SCHMIDT	DATE 29-APR-2004	APPR BY	CHECK	DATE		FIRST MADE FOR ZEUS UPS	
REV PCN	DESCRIPTION	APPR BY	OTHER	DATE		DRAWING NO. 2142192SCH	REV 1
MADE BY	DATE					SIZE 8	SHT/CONT. ON 1 / 2

2142192SCH

See the GEMS ePDM System to determine the status of this document.

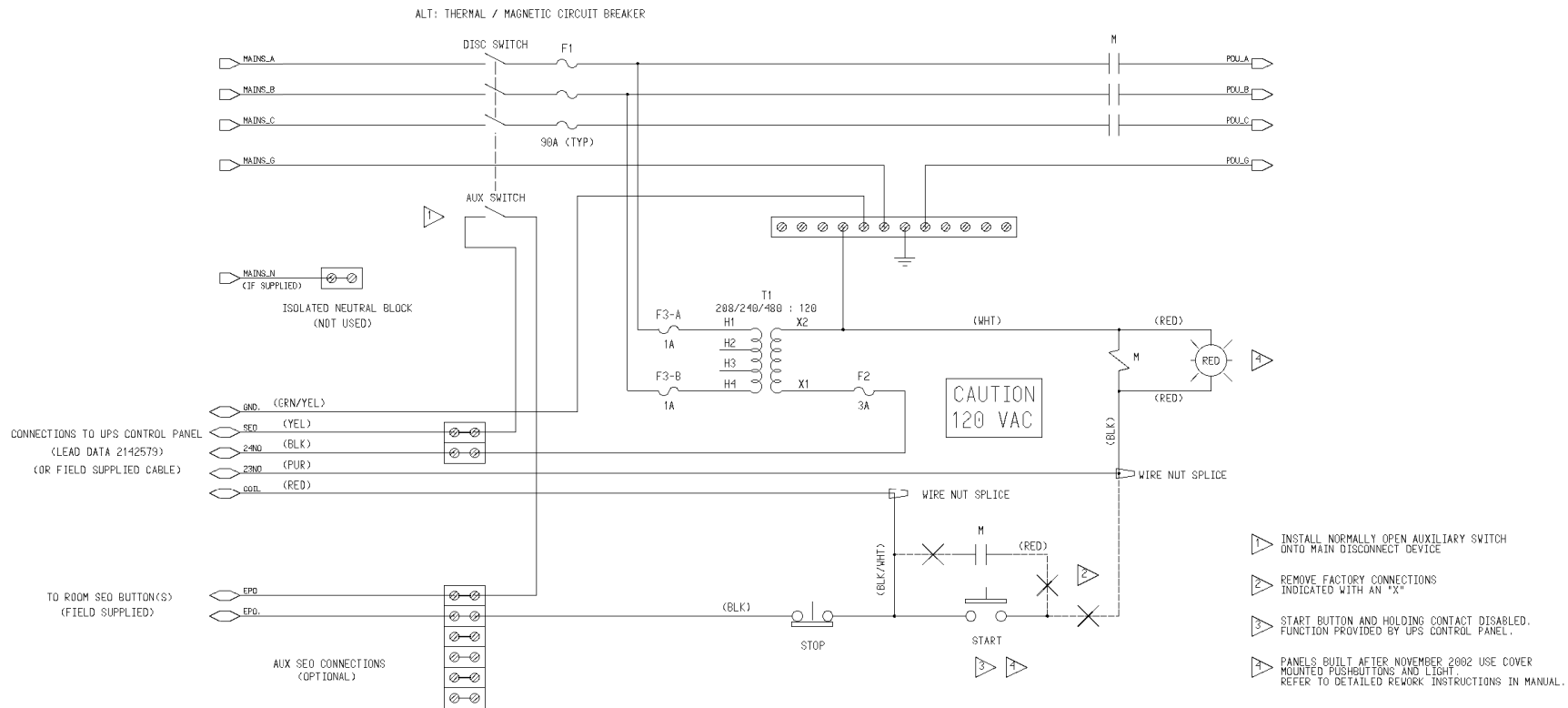
UPS INTERCONNECT



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MADE BY	J. R. SCHMIDT	DATE	29-APR-2004	APPR BY		CHECK		DATE		MILWAUKEE, WI USA	
REV	PCN		DESCRIPTION		DATE	APPR		DATE		DESIGN TITLE	UPS INTERCONNECT
MADE BY						OTHER				FIRST MADE FOR	ZEUS UPS
										DRAWING NO.	2142192SCH
										REV	1
										SIZE	B
										SHT/CONT.	ON 2 / 3

See the GEMS ePDM System to determine the status of this document.

2142192SCH

A1 (MAINS) DISCONNECT
R4502AD or EQUAL

BLOCK PATHNAME \$GEMS_CT_XRAY_GEN/INTERCONNECT/2142192_SCH/UPS_IC/A1.DISC

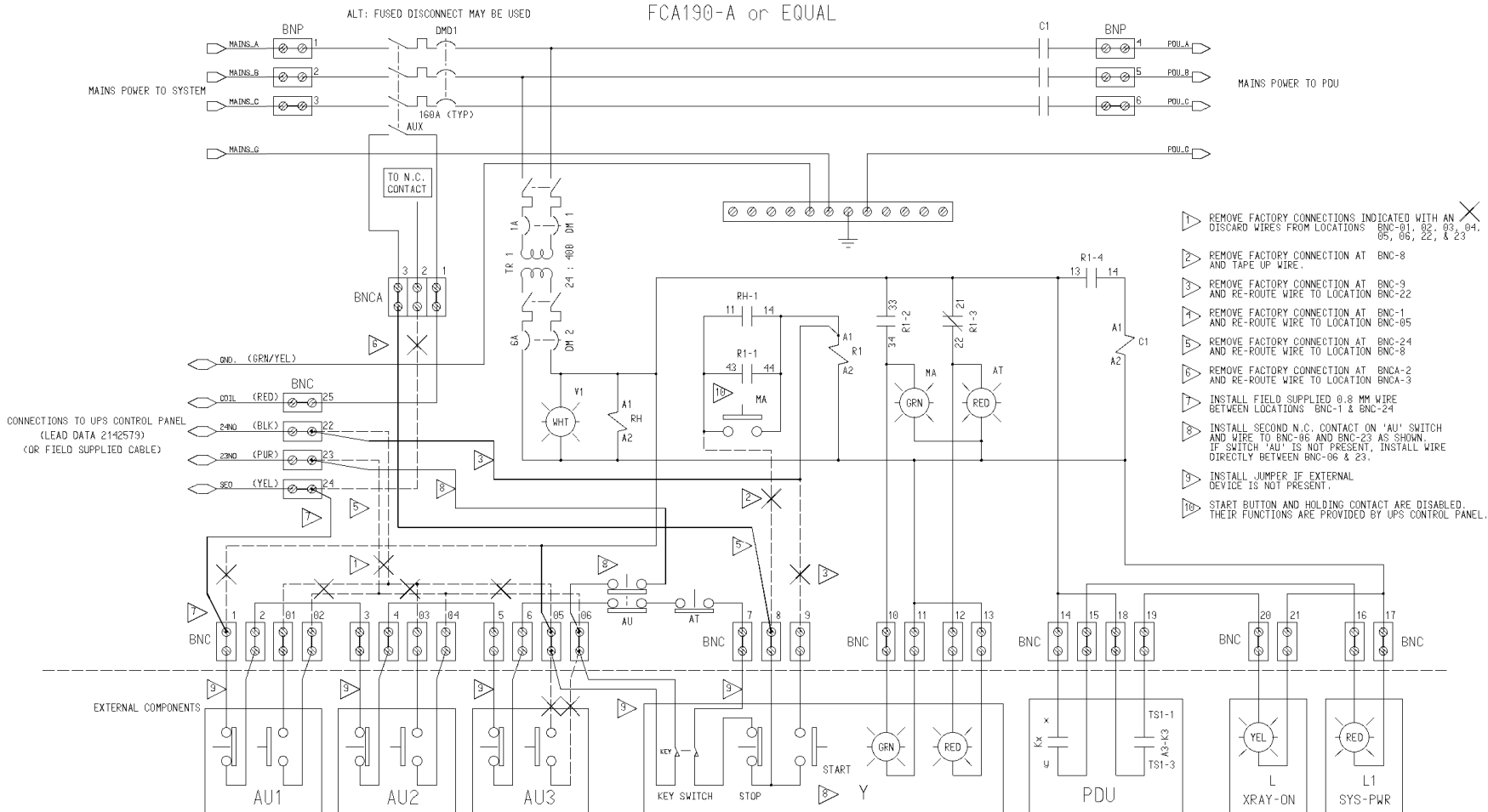
BLOCK SHT 1 OF 3

REV 1	PCN 2002189	DESCRIPTION ADDED INFO FOR NGPDU-2	MADE J. R. SCHMIDT	DATE 10/DEC/2000	GE MEDICAL SYSTEMS	DESIGN TITLE UPS INTERCONNECT	REV 1	SIZE B	SH/CONT. ON 3 / 4
MADE BY J. R. SCHMIDT	DATE 23-APR-2004	APPR BY	CHECK APPR OTHER	DATE	MILWAUKEE, WI USA	FIRST MADE FOR ZEUS UPS			
REV PCN	DESCRIPTION	DATE	APPR BY	DATE		DRAWING NO. 2142192SCH			

2142192SCH

See the GEMS ePDM System to determine the status of this document.

A1 (MAINS) DISCONNECT FCA190-A or EQUAL

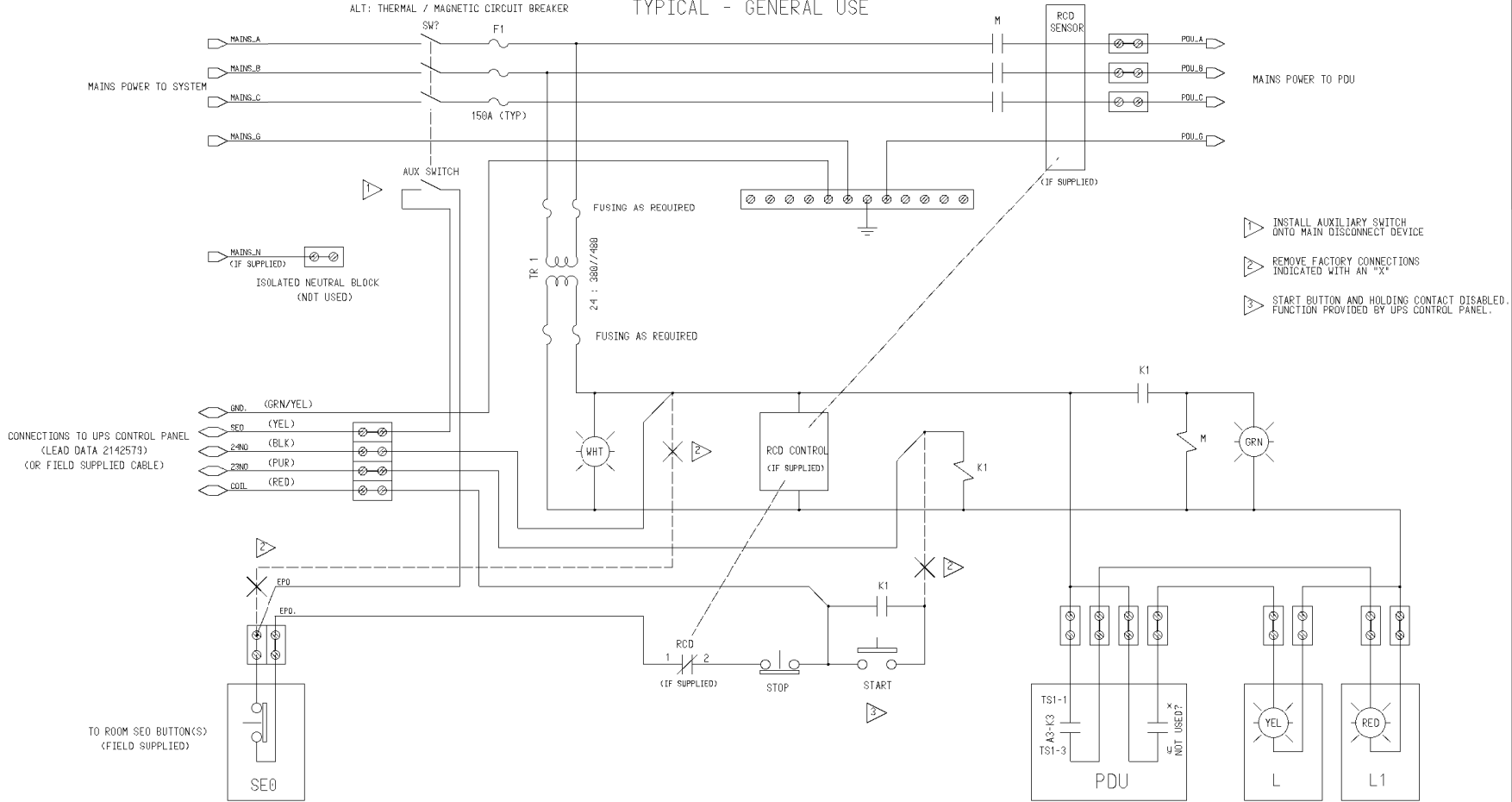


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REV 1	PCN	DESCRIPTION	DATE	APPR BY	DATE	MADE BY	DATE	DESIGN TITLE UPS INTERCONNECT FIRST MADE FOR ZEUS UPS	DRAWING NO. 2142192SCH	REV 1
MADE BY	PCN	DESCRIPTION	DATE	APPR BY	DATE	MADE BY	DATE	SIZE 8	SHT/CONT. ON 4 / 5	

See the GEMS ePDM System to determine the status of this document.

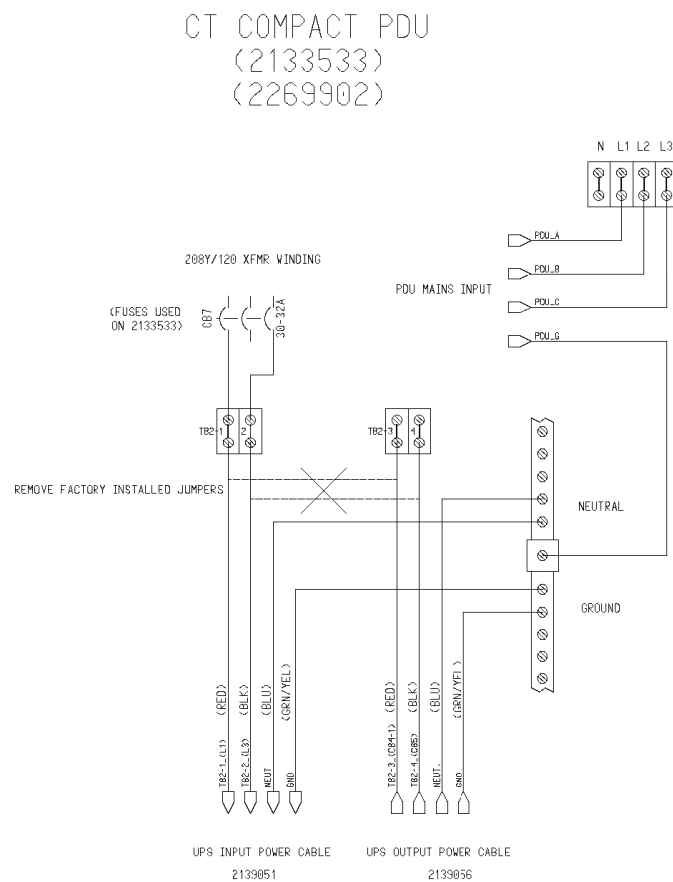
2142192SCH

A1 (MAINS) DISCONNECT TYPICAL - GENERAL USE



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REV 1	PCN 2002189	DESCRIPTION 2002189	DATE 23-APR-2004	APPR BY	DATE	DESIGN TITLE UPS INTERCONNECT	REV 1
MADE BY J. R. SCHMIDT	DATE 23-APR-2004	APPR BY	DATE	MADE CHECK APPR OTHER	DATE	FIRST MADE FOR ZEUS UPS	SIZE B
REV PCN	DESCRIPTION	DATE	APPR BY	DATE	DATE	DRAWING NO. 2142192SCH	SHT/CONT. ON 5 / 6
MADE BY	DATE	APPR BY	DATE	MADE CHECK APPR OTHER	DATE	GE MEDICAL SYSTEMS MILWAUKEE, WI USA	

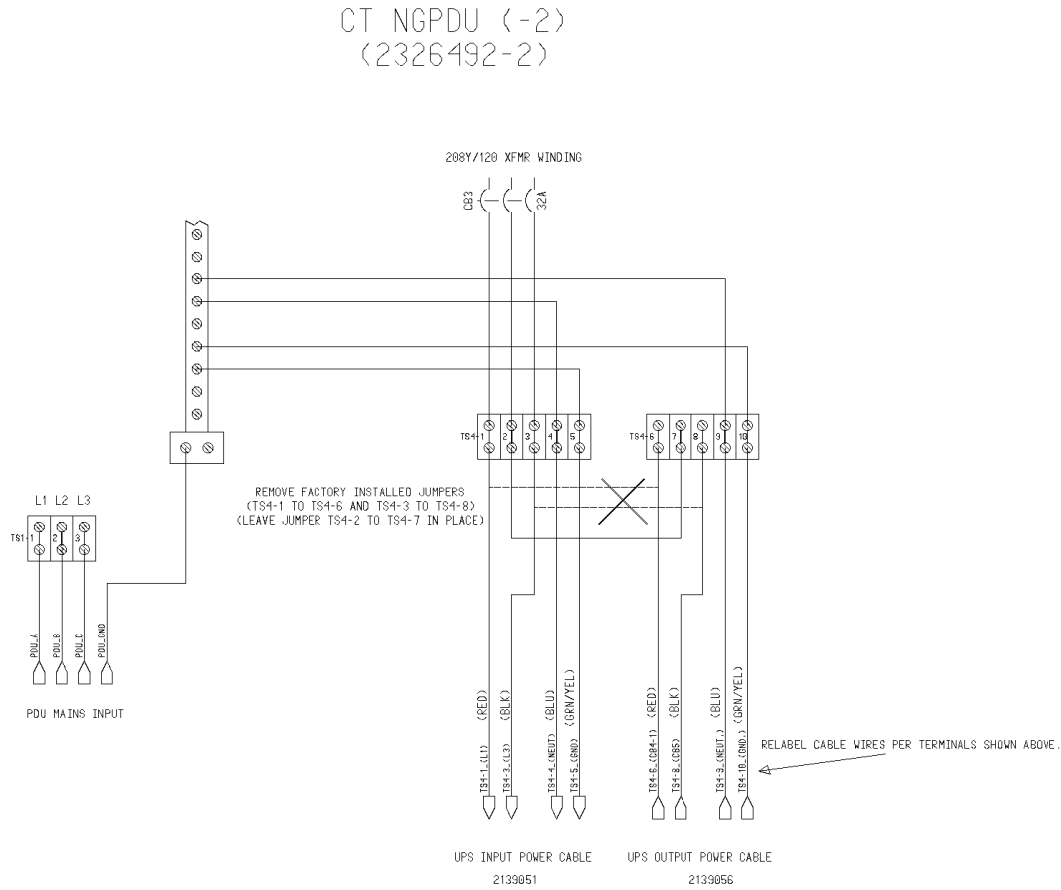
2142192SCH



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MADE BY	J. R. SCHMIDT	DATE	29-APR-2004	APPR BY		DATE						FIRST MADE FOR	ZEUS UPS			
REV	PCN		DESCRIPTION			APPR		DATE		MILWAUKEE, WI USA		DRAWING NO.		REV	SIZE	SHT/CONT. ON
MADE BY		DATE		APPR BY		DATE							2142192SCH	1	B	6 / 7

See the GEMS ePDM System to determine the status of this document.

2142192SCH

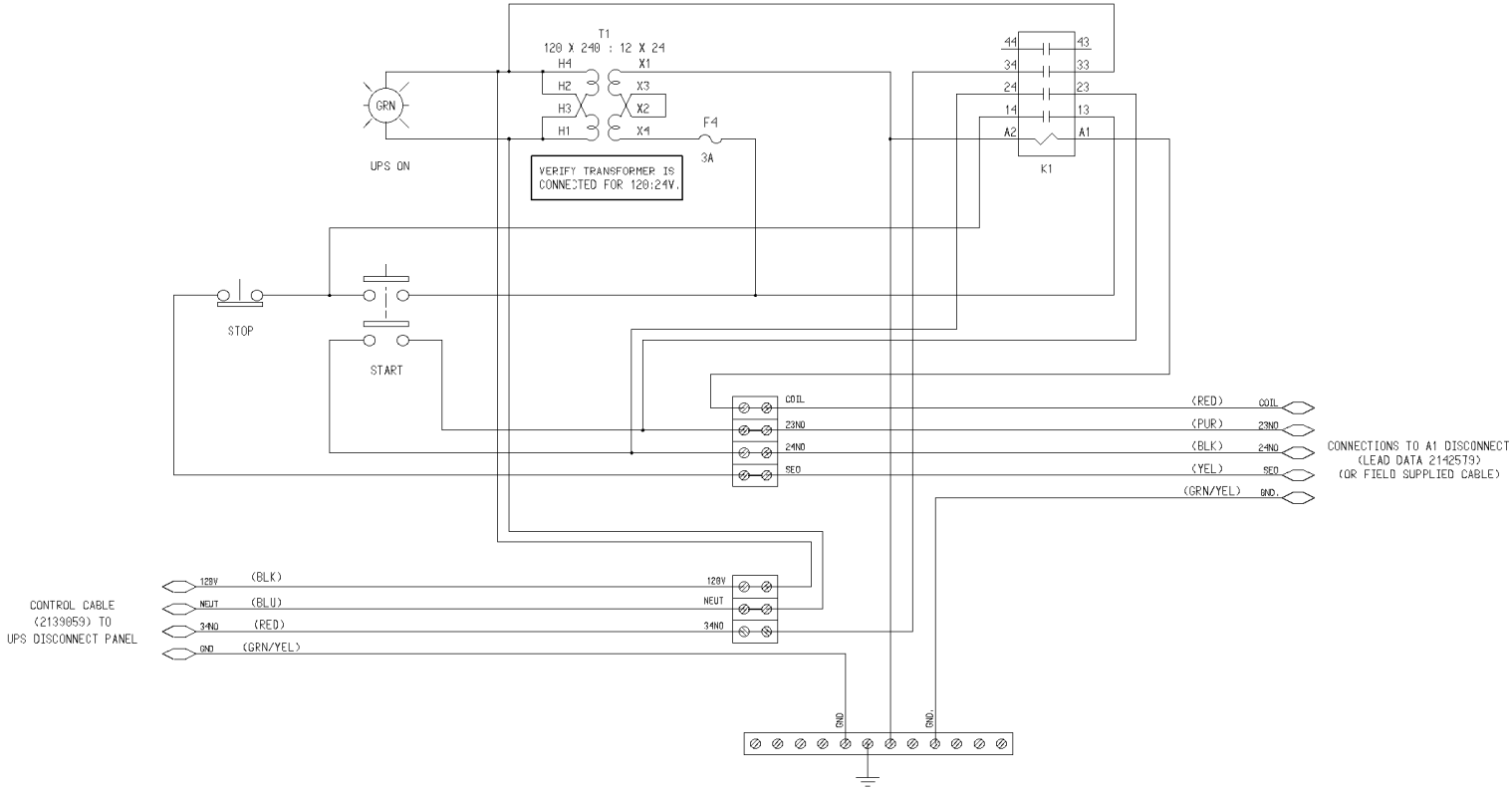


BLOCK PATHNAME \$GEMS_CT_XRAY_GEN/INTERCONNECT/2142192_SCH/UPS_IC/PDU					BLOCK SHT 2 OF 2				
REV 1	PCN 2002189	DESCRIPTION ADDED INFO FOR NGPDU-2	MADE J. R. SCHMIDT	DATE 10/DEC/2000	GE MEDICAL SYSTEMS MILWAUKEE, WI USA		DESIGN TITLE UPS INTERCONNECT		
MADE BY J. R. SCHMIDT	DATE 23-APR-2004	APPR BY	CHECK	DATE			FIRST MADE FOR ZEUS UPS		
REV	PCN	DESCRIPTION	MADE	DATE	MILWAUKEE, WI USA		DRAWING NO.		
MADE BY	DATE	APPR BY	APPR OTHER	DATE			2142192SCH		
							REV 1	SIZE 8	SHT/CONT. ON 7 / 8

2142192SCH

See the GEMS ePDM System to determine the status of this document.

UPS CONTROL PANEL
(2143207)

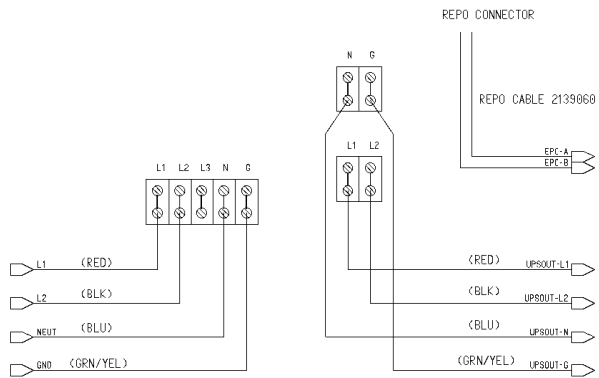


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MADE BY	J. R. SCHMIDT	DATE	29-APR-2004	APPR BY		CHECK		DATE		MILWAUKEE, WI USA	FIRST MADE FOR
REV	PCN	DESCRIPTION		APPR BY		OTHER		DATE			UPS INTERCONNECT
MADE BY		DATE									ZEUS UPS
										DRAWING NO.	2142192SCH
										REV	1
										SIZE	8
										SHT/CONT. ON	8 / 9

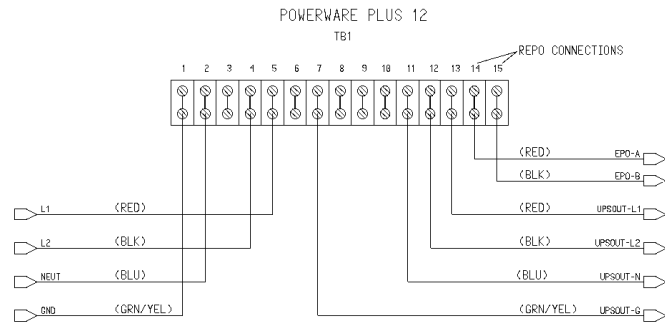
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2142192SCH

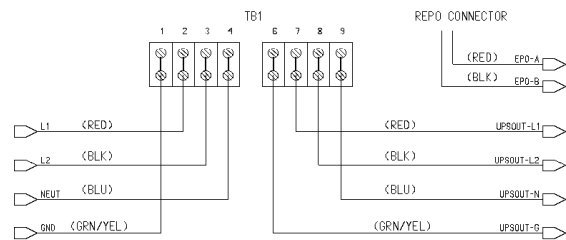
UPS (LIEBERT)



UPS (POWERWARE)



POWERWARE 9150 SERIES

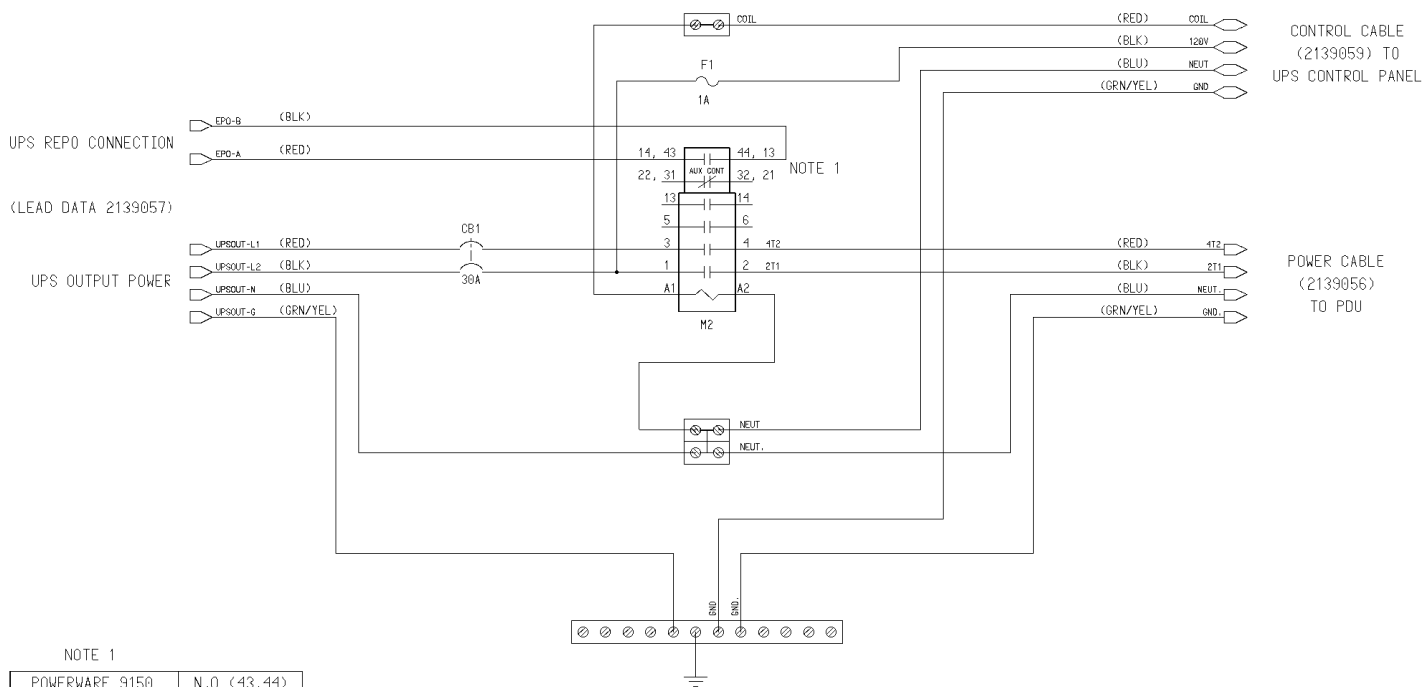


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MADE BY J. R. SCHMIDT	DATE 29-APR-2004	APPR BY			DATE	FIRST MADE FOR ZEUS UPS								
REV PCN	DESCRIPTION	APPR BY			DATE	DATE	DRAWING NO. 2142192SCH			REV 1	SIZE B	SHT/CONT. ON 9 / 10		
MADE BY	DATE	APPR BY			DATE	DATE								

2142192SCH

See the GEMS ePDM System to determine the status of this document.

UPS DISCONNECT PANEL (2143161)



NOTE 1

POWERWARE 9150	N.O (43,44)
POWERWARE PLUS 12	N.C (31,32)
POWERWARE PROFILE	N.O (43,44)
LIEBERT	N.C (31,32)

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MADE BY J. R. SCHMIDT		DATE 29-APR-2004	APPR BY	DATE		CHECK		MILWAUKEE, WI USA		FIRST MADE FOR ZEUS UPS			
REV PCN		DESCRIPTION	DATE	APPR BY	DATE	OTHER		DRAWING NO.	2142192SCH	REV 1	SIZE 8	SHT/CONT. ON 10 / -	

Section 2.0 - Installation (2142192IDW, Rev 3)

REVISIONS					
REV	QTY	MADE BY	CH	DATE	PCN/FZ2 NO.
1		JOE ROBINETTE		97DEC04	189597
RELOCATED EXIDE UPS NOTE					
2		TIM BEHLMER		98MAR05	199193
REVISED DWG. UPDATE VIEWS SHOWING INSTALLATION					
3		GENE GRUBER		01JUL31	220211
WAS SHEET 21 OF 28 - WAS EXIDE UPS ONLY - RED BLK OLU WAS BLK RED WHT AT TB1					

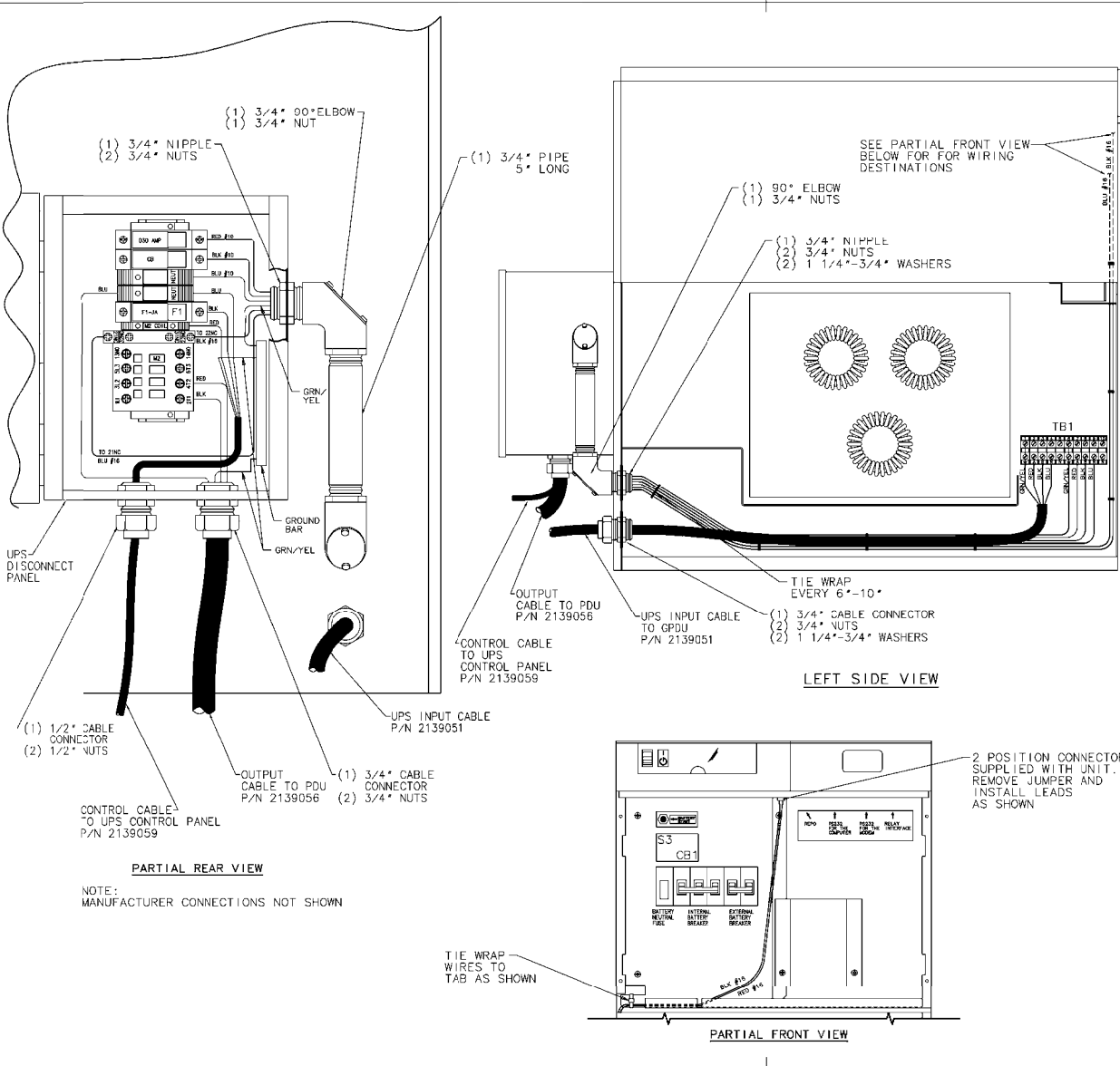
THE REVISION NUMBER OF THIS SHEET IS THE CONTROL NUMBER OF THIS ENTIRE DOCUMENT. THIS REVISION NUMBER MUST INCREASE WHENEVER ANY SHEET IS CHANGED. THE CHANGE MUST BE RECORDED IN THE FORM BELOW.

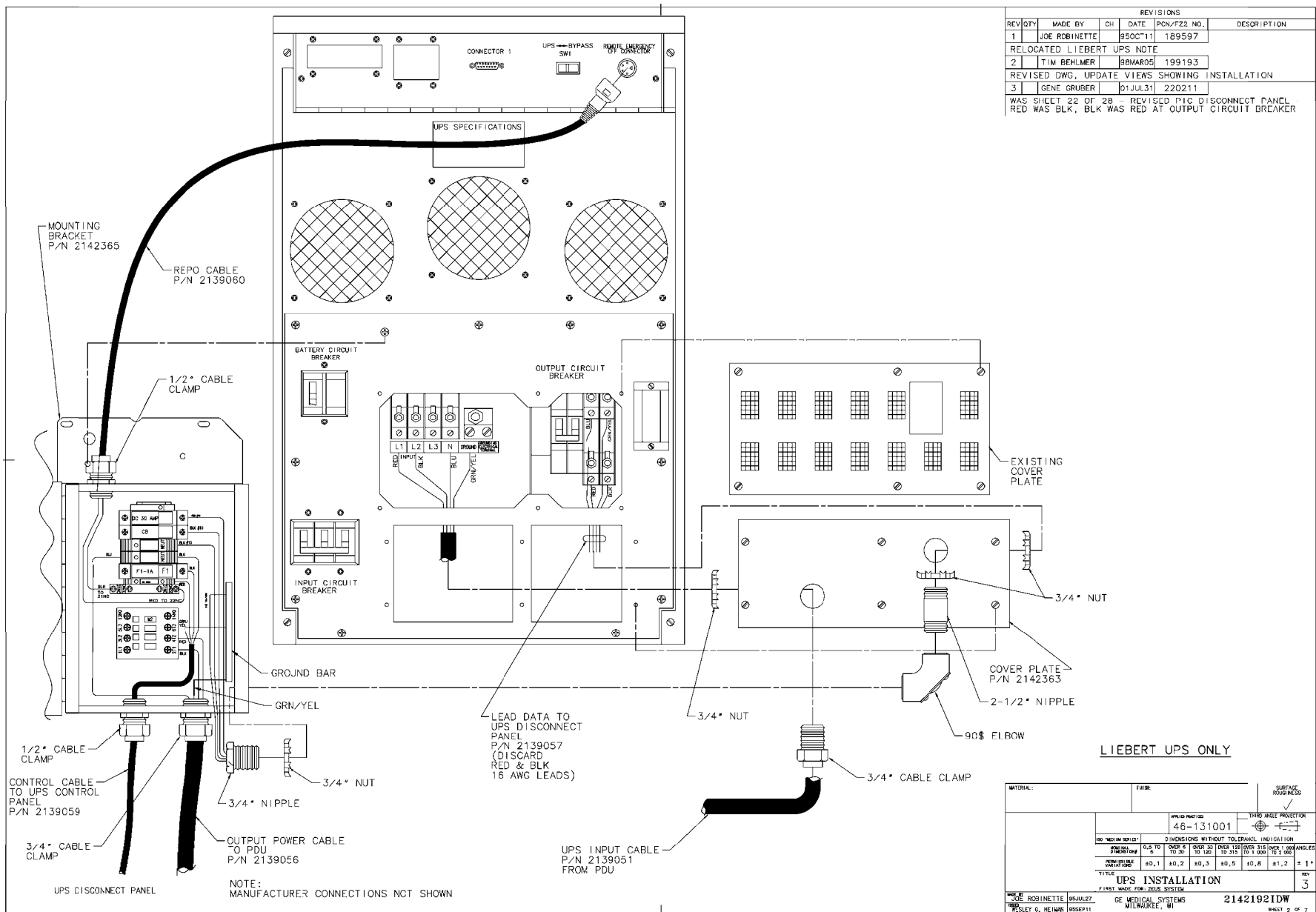
PCN #	DATE	BY	CHK	APP
220211	01/31/01	GENE GRUBER		
1				
2				
3				
4				
5				
6				
7				

NOTE:
1. FOR ELECTRICAL SCHEMATICS SEE 2142192SCH

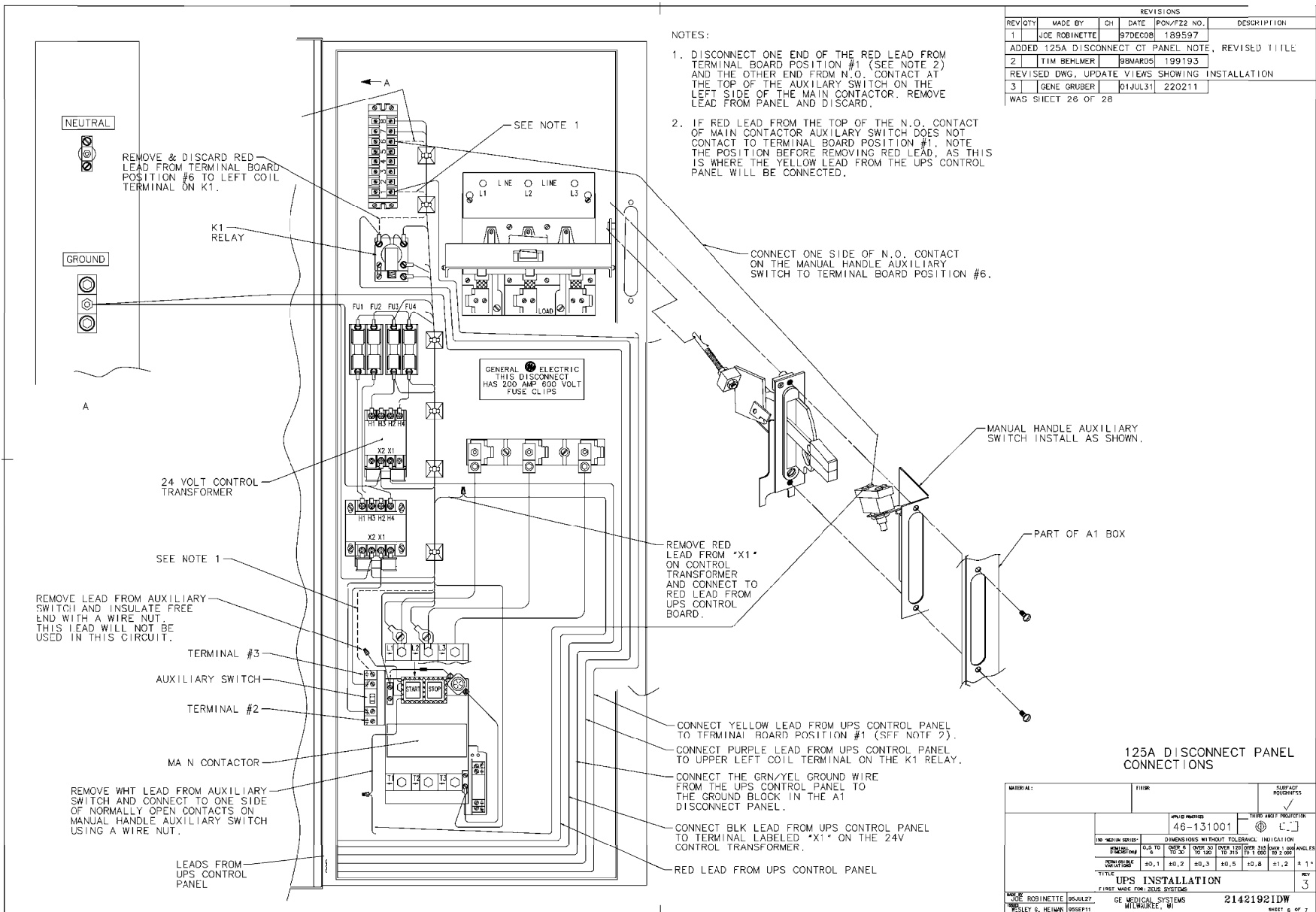
POWERWARE UPS ONLY

MATERIAL:		FIELD:		SURFACE PROTECTION:	
DRAWING: METRIC		46-131001		THESE ARE PROTECTED	
DATE: 8/20/95		DATE: 8/20/95		DATE: 8/20/95	
BY: MCH		BY: MCH		BY: MCH	
JOB: JR		JOB: JR		JOB: JR	
APPROVALS: 11/06/00		APPROVALS: 11/06/00		APPROVALS: 11/06/00	
JOE ROBINETTE		JOE ROBINETTE		JOE ROBINETTE	
WESLEY G. NEUMAN		WESLEY G. NEUMAN		WESLEY G. NEUMAN	
GE MEDICAL SYSTEMS		GE MEDICAL SYSTEMS		GE MEDICAL SYSTEMS	
MILWAUKEE, WI		MILWAUKEE, WI		MILWAUKEE, WI	
2142192IDW		2142192IDW		2142192IDW	
SHEET 1 OF 2		SHEET 1 OF 2		SHEET 1 OF 2	

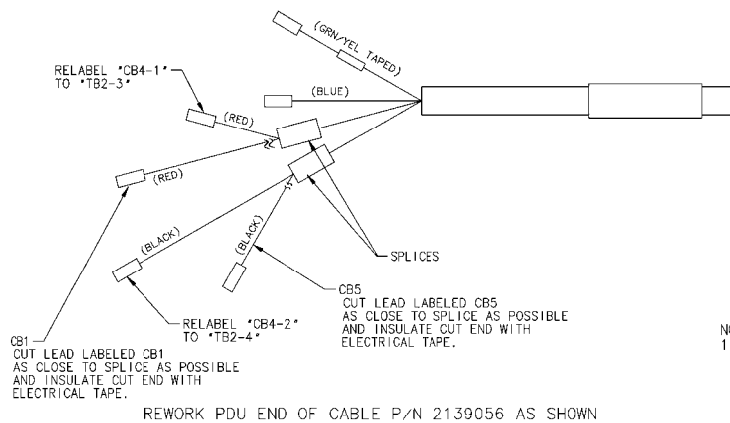
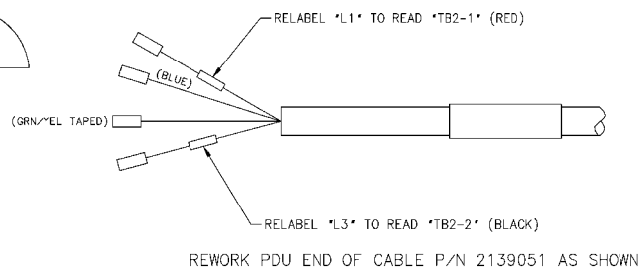








REVISIONS						
REV	DATE	BY	CHK	DATE	NO./72	DESCRIPTION
1	10/02/93	JOE ROBINETTE		9/24/93	N/A	
NOTE: REMOVE 2 EXISTING JUMPS FROM BOTTOM OF TB2 AND INSTALL CABLES						
2	11/01/93	TIM BEHLMER		09/24/93	199193	
REVISE DWG., UPDATE VIEWS SHOWING INSTALLATION						
3	01/03/94	GENE GRUBER		01/03/94	220211	
WAS SHEET 27 OF 28 - ADDED COMPACT PDU CONNECTIONS ADDED COLORS TO LEADS						



NOTES:
1. INSTALL THIS LEAD INTO ANY
OPEN POSITION

COMPACT PDU CONNECTIONS

[illegible]

Appendix A

Applicable GEMS Documents

The following documents form a part of this specification to the extent applicable. All documents are under revision control and any changes require a Product Change Notice (PCN). Where listed, document revisions are for information only. The latest revision of each document in effect at time of purchase shall apply.

Section 1.0

Controlling Documents

- 2142192SCH UPS Interconnect Schematic
- 2269902SCH H2 PDU Assembly - Schematic
- 2143207PSP Purchase Specification for UPS Control Panel
- 2143161PSP Purchase Specification for the UPS Disconnect Panel
- 2147944PSP Purchase Specification for the UPS Hardware Kit
- 46-015900 Global Documentation Standard
- 46-131001 Applied Practices (Assemblies)
- 46-170700 Applied Practices (Portions)

Section 2.0

Standards Documents

- 46-319010 Revision 0 Safety - Electrical power grounding 806-1
- 46-319011 Revision 0 Safety - Electrical leakage currents 806-2
- 46-319012 Revision 0 Safety Checklist for Design 806-3
- 46-319013 Revision 1 Operator Manual Safety/Regulatory Checklist 806-4
- 46-319015 Revision 0 Safety - Product Safety Labels and Instructions 806-6

Section 3.0

Procedure Documents

- POWERWARE 9150 User's Guide
- POWERWARE Plus 12 User's Guide
- LIEBERT UPS User's Guide

Section 4.0

Government, Safety and Regulatory Documents

- UL 2601 General Requirements for Safety
- UL 94 Tests for Flammability of Plastic Materials
- CSA 22.2 Canadian Standards Association
- IEC 601-1 General Requirements for Safety
- EN 60601-1-2 IEC Normalized EMC specification
- IEC 801-2 Electro-Static Discharge
- IEC 801-3 Radiated Electro-Magnetic Fields Requirements
- IEC 801-4 Electrical Fast Transient / Burst Requirements
- IEC 801-5 Surge Immunity Requirements
- IEC 801-6 Immunity to Conducted Disturbances, Induced by RF fields above 9 kHz.
- IEC 1000-4-8 Magnetic Fields

Section 5.0

Industry Standards

- NEMA XR-7
- NFPA-70 (1999) National Electrical Code (USA)

Appendix B

Field Replaceable Unit (FRU) Listing

The following listing identifies the Field Replaceable Units (FRUs) associated with this assembly. Vendor part numbers are included as examples only. For the complete list of approved vendors for each item, see the associated ESL on GEMS GPCP system.

GEMS PART #	NAME /DESCRIPTION	VENDOR	VENDOR P / N
2139965	UPS	POWERWARE	PF1221114UE0000
2139965-2	UPS	LIEBERT	VUB12CBGA
2143161	UPS Disconnect Panel	GE Supply	2143161
2143207	UPS Control Panel	GE Supply	2143207
2147944	Hardware Kit	GE Supply	2147944

Table B-1 Field Replaceable Units



GE MEDICAL SYSTEMS

***3000 N, GRANDVIEW BLVD,
WAUKESHA, WI 53188, USA.***